

Block 6

Operational Aspects of Financial Markets

Unit 18	Efficient Portfolio Frontier
Unit 19	Capital Asset Pricing Model
Unit 20	Arbitrage Pricing Theory
Unit 21	Pricing of Derivatives
Unit 22	Corporate Finance
Unit 23	Foreign Direct Investment and Foreign Portfolio Investment

UNIT 18 EFFICIENT PORTFOLIO FRONTIER

Structure

18.0 Objectives

18.1 Introduction

18.2 Portfolio Management

18.2.1 Stages of Portfolio Management

18.2.2 Benefits of Portfolio Management

18.3 Relationship between Risk and Return

18.3.1 Risks Involved in Investments

18.3.2 Measuring Risks

18.3.3 Expected Returns

18.4 Valuation of Portfolio and Expected Returns from a Portfolio

18.4.1 Valuation of Portfolio

18.4.2 Expected Return and Risk from a Portfolio

18.5 Markowitz Portfolio Theory

18.5.1 Assumptions of the Markowitz Model

18.5.2 Opportunities Set/ Feasible Set of Portfolios

18.5.3 Efficient Set of Portfolios

18.5.4 Criticisms of the Efficient Frontier

18.6 Let Us Sum Up

18.7 Key Words

18.8 Some Useful Books and References

18.9 Answers/Hints to Check Your Progress Exercises

18.0 OBJECTIVES

After going through this unit, you will be able to:

- describe the importance of diversification;
- explain the need for portfolio management;
- calculate the expected return and valuation of the portfolio;
- explain the relationship between risk and return;
- critically examine Markowitz's Portfolio theory; and
- explain Efficient Portfolio frontier.

18.1 INTRODUCTION

In the realm of finance, the adage “Don't put all your eggs in one basket” holds significant importance. This principle underscores the fundamental strategy of spreading investments across various assets and asset classes to mitigate risk and optimize returns.

Diversification is the practice of spreading investments across various assets and asset classes to mitigate risk and optimize returns. It means creating a well-balanced portfolio, wherein a range of investments is judiciously selected and blended to counteract the impact of volatility in any single investment.

The need for diversification stems from the volatility inherent in financial markets. Given that certain investments may thrive while others may decline due to various factors such as economic recessions or geopolitical unrest, diversification serves as a strategy for investors to mitigate the impact of adverse market fluctuations. By spreading their investments across different assets, investors seek to protect their wealth from significant losses in the long run.

Diversification is a key principle within portfolio management. A portfolio represents a collection of investments held by an individual or entity, comprising a broad range of assets including stocks, bonds, mutual funds, real estate, and other financial instruments.

Many people find it difficult to make effective investment decisions because of uncertainty in the market, conflicting investment goals and sometimes even due to heavy information load. In response to these challenges, numerous theories in portfolio management have endeavoured to provide solutions. They have tried to answer questions like how to diversify. Which financial assets to invest in? How to combine the assets into a portfolio? What is the portfolio's risk? how to calculate the expected return? etc.

In this unit, we will discuss portfolio management, expected return and valuation of portfolio, relationship between risk and return. We will also discuss Markowitz's Portfolio theory and Efficient Portfolio frontier. In subsequent units, we will talk about the Capital asset pricing model (CAPM) and Arbitrage pricing theory (APT).

18.2 PORTFOLIO MANAGEMENT

A portfolio is essentially a collection of investments owned by an individual or entity, which can include various types of assets like stocks, bonds, mutual funds, and real estate. However, it's more than just a bunch of investments grouped; it's a strategic approach to managing wealth, minimizing risks, and achieving financial goals.

Portfolio management is the practice of overseeing and optimizing these investment collections. It involves carefully deciding how to allocate resources, balancing risk and returns, and adapting to changes in the market. This includes activities such as deciding which assets to invest in, evaluating the level of risk involved, and monitoring how well the portfolio is performing.

The main goal of portfolio management is to create portfolios that match the specific financial goals, risk tolerance, and timeframes of investors. Whether someone wants to preserve capital, generate income, or increase wealth, portfolio managers develop strategies tailored to these objectives. They do this through thorough analysis and smart decision-making, aiming to achieve the best possible returns while minimizing the chances of losses.

Additionally, portfolio management is an ongoing process that requires constant attention and adjustment. Changes in the market shifts in the economy, and alterations in investors' situations mean that portfolios need to be reviewed and adjusted regularly. This means that successful portfolio management relies on a combination of financial expertise, careful research, and proactive decision-making.

18.2.1 Stages of Portfolio Management

Portfolio management is a dynamic procedure that includes the following fundamental stages:

- Understanding the goals, constraints, and preferences of investors to create an investment policy.
- Creating and executing strategies aligned with the formulated investment policy.
- Choosing asset classes and specific securities within each class based on their risk-return characteristics.
- Regularly assessing and tracking the portfolio's performance by consistently evaluating market conditions and company performance.
- Analyzing the portfolio's outcomes to compare against predetermined targets and making necessary adjustments for future enhancements.

18.2.2 Benefits of Portfolio Management

- 1) **Consistent Risk Measurement:** It ensures that risks are measured in the same way every time, providing a clear understanding of potential risks.
- 2) **Considering Correlation and Diversification:** It takes into account how different parts of the organization or investment options relate to each other. This understanding can help in reducing overall risk.

- 3) **Guidance for Strategic Planning:** It provides direction for the company's plans, highlighting areas where improvements are needed in terms of risk and return.
- 4) **Increased Transparency in Decision Making:** Portfolio management helps make decisions clearer by consistently evaluating all business units and options.
- 5) **Understanding Value Creation:** Portfolio management helps in better grasping how new investment opportunities can create value for the organization.
- 6) **Identifying Improvement Areas:** Portfolio management helps identify where the company needs to focus to enhance its risk-return position.
- 7) **Considering Non-Financial Implications:** It acknowledges the qualitative and non-financial impacts of decisions, not just the financial aspects.

Check Your Progress 1

- 1) What is portfolio management? What are the different stages of portfolio management?
.....
.....
.....
- 2) Why is portfolio management important for an investor?
.....
.....
.....

18.3 RELATIONSHIP BETWEEN RISK AND RETURN

Most investors are risk-averse. This does not mean that people will not take a risk. That means they will take a risk only if they expect to be adequately rewarded for taking it. People have different degrees of risk aversion. Some are more willing to take a chance than others. Choosing an investment management strategy involves weighing the trade-off between risk and return. Active management might yield higher returns but comes with a greater risk of not meeting the portfolio's required return. Conversely, passive management offers market-based returns in diversified portfolios, reducing risky bets but potentially limiting excess returns. When deciding between passive and active management, it's crucial to determine if the goal is to solve a cash flow problem over time or to outperform the market.

18.3.1 Risks Involved in Investments

Total risk refers to the overall variability of the returns of a financial asset. It combines both diversifiable and non-diversifiable risks and represents the overall uncertainty and potential for loss associated with an investment. Recognizing the types of risks allows investors to implement appropriate risk management strategies. Understanding the different types of risks helps investors construct well-balanced portfolios. By diversifying across assets and sectors, investors can reduce the impact of specific risks on their overall portfolio performance.

The following types of risks are faced in the market:

Diversifiable risk, also known as unsystematic risk (unique risks), refers to firm-specific risks that affect individual stock prices rather than the entire industry or sector.

The key components of diversifiable risk are business, financial, and management risks. Diversifiable risk is crucial for investors seeking better returns and protecting their initial investment.

An uncertain risk can arise from various factors such as scams, labour strikes, regulatory penalties, management changes, internal issues, or news specific to the firm. Diversifiable risk can be mitigated through diversification, which involves spreading investments across different assets or sectors. By diversifying, investors can reduce the impact of adverse events on any single investment.

Systematic risks, or non-diversifiable risks, are the risks that affect the overall market or a specific sector. It is influenced by factors that impact the entire economy, such as macroeconomic indicators, market sentiment, or geopolitical events. Systematic risks are inherent risks that exist in the stock market. They're also called "non-diversifiable risks" or "market risks" since they impact the entire asset class.

Non-diversifiable means that an organization can't control, minimize, or avoid systematic risks. These risks are typically due to various external factors like the current geopolitical situation, monetary policy, and natural disasters. Non-diversifiable risk cannot be eliminated through diversification because it is inherent in the market itself. Investors are exposed to non-diversifiable risk regardless of how diversified their portfolios are. The only way to manage non-diversifiable risk is through strategies such as asset allocation or hedging. For example, COVID-19 is a systematic risk because it impacted the entire stock market.

Diversifiable risk analysis is essential for investors and portfolio managers as it helps in understanding and managing the risks associated with individual securities or assets. By assessing and diversifying their investments, investors can reduce the impact of firm-specific risks and potentially improve their risk-return profile. Diversifiable risk analysis enables investors to make

informed decisions about portfolio allocation, asset selection, and risk management strategies to achieve their investment objectives while mitigating unnecessary risks.

18.3.2 Measuring Risks

The risk linked to a stock pertains to the fluctuation in its rate of return. Investors need to quantify and assess this risk accurately. The risk probability distribution, depicting potential returns on investments, represents the total risk associated with an investment.

Variance

Variance takes into account the derivation of all possible returns from their mean or expected value. the formula used for variance is as follows:

$$\sigma^2 = \sum_{i=1}^n [R_i - E(R)]^2 X P$$

Where R_i refers to possible returns when $i=1$ to n , $E(R)$ is the expected value of the investment and p is the probability of the return.

Standard deviation can be calculated as the square root of variance, i.e.:

$$\sigma = \sqrt{\sum_{i=1}^n [R_i - E(R)]^2 X P}$$

18.3.3 Expected Returns

Expected return denotes the anticipated profit or loss an investor can reasonably foresee from an investment within a specific timeframe. It is an important factor that determines portfolio selection. To ascertain return, various factors like market trends, economic predictions and historical data are analysed.

The concept of expected return is useful because there is uncertainty about the return on investment. Thus, an expected return means the average return that an investor expects to receive on investment over some time. It is calculated by the sum of each possible return multiplied by its probability. The expected return is calculated by using the following formula:

$$E(R) = \sum_{i=1}^n R_i P_i$$

Where:

$E(R)$ =expected return on an investment

n =number of possible terms

R_i = the i^{th} possible return

P_i = the probability of i^{th} return R_i

Example 18.1: Mr X has invested an Equal amount in Security A and Security B. The expected return during boom and depression with equal probability of occurrence is as follows:

Economic condition	Expected return	
	Security A	Security B
Boom	15	12
Depression	5	7

Calculate the Expected return and standard deviation of each security.

Solution: Calculation of expected rate of return

Economic condition	Expected return of		Probability	Expected returns	
	Security A	Security B		Security A	Security B
Boom	15	12	0.5	7.5	6
Depression	5	7	0.5	2.5	3.5
Expected return (ER)				11.00	9.5

Calculation of standard deviation of Security A

Economic condition	R_A	$R_A - E(R)$	$(R_A - E(R))^2$	Probability	$(R_A - E(R))^2 \times P$
Boom	15	4	16	0.5	8
Depression	5	-6	36	0.5	18
Variance					26

Standard deviation of security A = $\sqrt{26} = 5.09$

Calculation of standard deviation of Security B

Economic condition	R_B	$R_B - E(R)$	$(R_B - E(R))^2$	Probability	$(R_B - E(R))^2 \times P$
Boom	12	2.5	6.25	0.5	3.125
Depression	7	-2.5	6.25	0.5	3.125
Variance					6.25

Standard deviation of security B = $\sqrt{6.25} = 2.5$

- 1) Explain the relationship between risk and return.

.....

- 2) What are diversifiable and non-diversifiable risks?

.....

18.4 VALUATION OF PORTFOLIO AND EXPECTED RETURNS FROM A PORTFOLIO

As we have discussed earlier, a portfolio is a collection of financial assets like stocks, bonds, mutual funds, exchange-traded funds (ETFs), real estate, commodities, and other investment vehicles owned by an individual, institution, or entity. The purpose of building a portfolio is to diversify risk and potentially maximize returns by spreading investments across different asset classes, industries, and geographic regions. The expected return and valuation of the Portfolio act as guiding principles and significantly influence investment choices and the assessment of performance.

18.4.1 Valuation of Portfolio

Portfolio valuation entails evaluating the value of an entire investment portfolio at a point in time. This process includes assessing the worth of each asset within the portfolio, taking into account various factors like market prices, potential earnings, and intrinsic value. Precise portfolio valuation is essential for evaluating performance, making well-founded investment choices, and ensuring transparency with stakeholders.

Let a portfolio be a collection of n assets, each with share price P_i ($i=1,2,3,\dots,n$) Let there be N_i shares of each asset i .

Then, the total value of the portfolio will be:

$$V = N_1P_1 + N_2P_2 + N_3P_3 + \dots \dots N_nP_n = \sum_{i=1}^n N_iP_i$$

A typical portfolio has $V > 0$ with the weight of each investment as:

$$w_i = \frac{N_iP_i}{N_1P_1 + N_2P_2 + \dots \dots N_nP_n} = \frac{N_iP_i}{V}$$

Factors affecting portfolio valuation:

- 1) *Asset Evaluation*: Determining the value of individual assets within a portfolio involves the valuation of each asset class. Some of the evaluation methods are discounted cash flow analysis, price-to-earnings ratios, or comparative company analysis.
- 2) *Market Dynamics*: Fluctuations in market conditions and liquidity constraints can impact asset prices, thereby influencing portfolio valuation. Regular monitoring of market trends ensures that portfolio valuations accurately reflect prevailing market conditions.
- 3) *Impact of Diversification*: The benefits of diversification inherent in a portfolio can affect its valuation. Diversified portfolios may display reduced volatility and higher valuation metrics owing to the mitigation of risk.

18.4.2 Expected Return and Risk from a Portfolio

The portfolio return is a weighted average of the individual returns.

Consider a portfolio of n assets with weights: $\{w_1, w_2, w_3, w_4 \dots w_n\}$, $\sum_{i=1}^n w_i = 1$

Then, the return on the portfolio will be:

$$\tilde{r}_p = w_1\tilde{r}_1 + w_2\tilde{r}_2 + \dots \dots w_n\tilde{r}_n$$

Where w_i is the weight of each asset and $\tilde{r}_i$ is a return from each investment

The expected return on the portfolio is:

$$\bar{r}_p = E[r_p] = w_1\bar{r}_1 + w_2\bar{r}_2 + \dots w_n\bar{r}_n$$

Where w_i is the weight of each asset and $\bar{r}_i$ is the expected return from each investment

Risk

The variance and standard deviation are used as a measure of risk. According to modern portfolio theory, unlike the expected return, the variance of a portfolio is not a simple weighted average of the individual variances. Rather, it is affected by the correlation coefficient of securities also.

The general form of the relationship for an n security portfolio is:

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n x_i x_j \rho_{ij} \sigma_i \sigma_j$$

Where σ_p^2 = portfolio variance

x_i = proportion of total investment in asset i

x_j = proportion of total investment in asset j

ρ_{ij} = correlation in the returns of securities i and j

σ_i = security standard deviation

Let us understand it with the help of a two-security case:

$$\sigma_p^2 = x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2x_A x_B \rho_{AB} \sigma_A \sigma_B$$

All else being equal, risk is something people want to avoid. One measure of risk is portfolio variance. It represents total risk. Intuitively, most investors want portfolio variance to be as low as possible without having to give up any return. In the above equation, the total risk (σ_p^2) has three components: one from the variance of stock A, another from the variance of stock B, and a third from the relationship between returns of Stock A and Stock B i.e., interactive risk.

As the number of securities in the portfolio increases, the number of terms on the right-hand side of the equation increases as well. With four securities, there will be four variance components and eight interaction components from the covariance matrix.

The portfolio manager does not have any control over individual securities but has control over the selection of portfolios. They can try to choose securities which have the least correlation between them so that interaction terms can be reduced and total risk be minimised.

18.5 MARKOWITZ PORTFOLIO THEORY

Traditionally, portfolio theory focused on analyzing individual securities while considering the balance between risk and return. This theory operated on the assumption that securities should be selected based on minimizing risk, measured by their standard deviation from the mean of expected returns. According to this approach, there's a direct relationship between the variability of returns and risk: the higher the variability, the higher the risk, and vice versa. Therefore, investors typically opt for assets with the lowest variability of returns. To determine the return on an individual security, factors such as the company's dividend payments, price-earnings ratio, the duration of holding, and the market value of shares are considered.

Modern Portfolio Theory (MPT), a hypothesis put forth by Harry Markowitz in his paper "Portfolio Selection," (published in 1952 by the Journal of Finance) opened the door to modern portfolio theory. It is an investment theory based on the idea that risk-averse investors can construct portfolios to optimize or maximize expected return based on a given level of market risk, emphasizing that risk is an inherent part of higher reward. The modern portfolio theory emphasises the need for maximization of returns through a combination of securities whose total variability is lower. It is not necessary that the success could be achieved by trying to get all securities of minimum

risk. This was the first major publication indicating the importance of security return correlation in the construction of a stock portfolio. Markowitz showed that for a given level of expected return and a given security universe. (The entire collection of alternatives from which specific investment choices are made), finding the specific portfolio that dominates the others requires knowledge of the covariance or correlation matrix between all possible security combinations.

After the publication of this paper, numerous investment firms and portfolio managers began to program Markowitz algorithms, which prescribed portfolio proportions to minimize portfolio variants. Even today, the term Markowitz Diversification refers to portfolio construction accomplished using security covariances.

18.5.1 Assumptions of the Markowitz Model

- 1) Investors are rational.
- 2) The supply and demand equilibrium is instantly achieved.
- 3) There are no arbitrage opportunities.
- 4) Price moves are efficient
- 5) Access to information is available to all participants.
- 6) The market is liquid.
- 7) There is no transaction cost.
- 8) There are no taxes.
- 9) Everyone has the same opportunity of borrowing and lending.

The model is also known as the mean-variance model because it is based on the expected returns and the standard deviation of different portfolios. It helps to make the most efficient selection by analysing various portfolios of the given assets. It shows investors how to reduce the risk in case they have chosen assets, not moving together.

Under the assumptions, a single asset or portfolio of assets is considered to be efficient. If no other asset or portfolio of assets offers higher expected returns with the same or lower risk or lower risk with the same or higher expected Return.

18.5.2 Opportunities set/ Feasible set of Portfolios

With a limited number of securities, an investor can create numerous portfolios by combining these securities in various proportions. These combinations form what's called the portfolio opportunities set, representing the feasible options for investment. Each portfolio in this set is defined by its

expected return and a measure of risk, typically variance or standard deviation of returns.

Not every portfolio in the opportunities set is appealing to an investor. Some portfolios are superior to others. A portfolio dominates another if it either offers a lower standard deviation with the same expected return, or a higher expected return with the same standard deviation. Inefficient portfolios, which are dominated by others, are not of interest to investors. They focus instead on efficient portfolios, which offer the best risk-return trade-off.

18.5.3 Efficient Portfolio Frontier

Harry Markowitz initially conceptualized portfolio theory by defining the objective as identifying a collection of efficient portfolios. An efficient portfolio is one where no higher expected return can be attained without accepting a higher level of risk. Conversely, it's a portfolio where no greater certainty of returns can be obtained without sacrificing expected returns. Once this efficient set of portfolios is determined at the asset class level, the investment manager selects an allocation along that frontier that aligns most closely with the investor's desired risk-return profile for the fund. This collection of efficient portfolios is commonly referred to as the efficient frontier. Figure 18.1 illustrates a typical representation of such efficient portfolios.

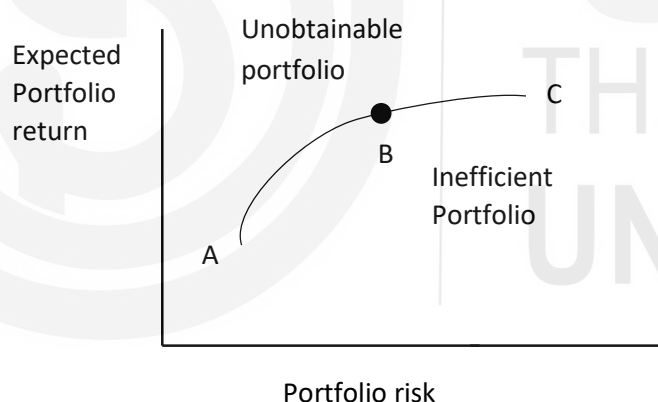


Fig. 18.1: Efficient Portfolio Frontier

This chart depicts the relationship between the expected return and the level of risk associated with a portfolio, or conversely, the risk of a portfolio about its expected return. The curved line shown in Figure 18.1 represents the efficient frontier, outlining the boundary of all possible portfolios formed by various combinations of selected asset classes. Portfolios lying inside this frontier are deemed inefficient because they offer less return for a given level of risk. Conversely, portfolios lying outside the line are more desirable, but they are not feasible to achieve.

Portfolio A represents the portfolio with the minimum risk, providing the least amount of risk among all possible combinations of asset classes. On the other hand, Portfolio C represents the portfolio with the maximum return and

the maximum risk. Although Portfolio C entails a 100% investment in the asset class offering the highest return, Portfolio A typically won't consist solely of the least risky asset class. This is because diversification across asset classes with varying risks helps mitigate overall risk. For portfolios seeking a balance between return and risk, options such as Portfolio B, positioned away from the endpoints, are usually preferred.

18.5.4 Criticisms of Markowitz's Portfolio Theory

While Markowitz's portfolio theory provides a valuable framework for portfolio construction, it is not without its limitations and criticisms. Some of the criticisms are:

- 1) **Unrealistic assumptions:** Markowitz's theory relies on several assumptions that might not hold in real-world scenarios. For instance, it assumes that investors are rational, risk-averse, and have access to unlimited borrowing and lending at a risk-free rate. However, in reality, investors may not always behave rationally, and their risk preferences can vary.
- 2) **Complexity and Practicality:** Implementing Markowitz's theory can be challenging due to its complexity. It involves intricate calculations and assumptions about correlations and expected returns, which may not always be accurate or practical to estimate.
- 3) **Sensitivity to Input Data:** The efficiency of portfolios generated by Markowitz's theory is highly sensitive to input data such as expected returns and covariance matrices. Small changes in these inputs can lead to significantly different portfolio allocations, making the results unstable and difficult to rely on.
- 4) **Neglect of Transaction Costs and Taxes:** Markowitz's theory assumes there are no transaction costs and taxes associated with portfolio rebalancing. In real-world investing, these costs can significantly impact returns and portfolio performance.
- 5) **Single-Period Analysis:** The theory primarily focuses on single-period analysis, neglecting the dynamic nature of markets and investors' changing risk preferences over time. This limitation can lead to suboptimal portfolio decisions in dynamic market conditions.
- 6) **Ignorance of Behavioural Finance Insights:** Markowitz's theory largely ignores insights from behavioural finance, which studies how psychological biases influence investor behaviour. Behavioural factors such as herd mentality, overconfidence, and loss aversion can significantly affect portfolio decisions but are not adequately accounted for in traditional portfolio optimization frameworks.
- 7) **Inability to Capture Tail Risks:** Traditional portfolio optimization techniques, including Markowitz's theory, often fail to adequately

capture extreme events or tail risks, leading to underestimation of potential portfolio losses during market downturns.

Check Your Progress 3

- 1) What are the key points of Markowitz's portfolio theory?

.....

.....

.....

- 2) What is an Efficient Portfolio Frontier?

.....

.....

.....

- 3) Discuss some of the limitations of Markowitz's portfolio theory.

.....

.....

.....

.....

18.6 LET US SUM UP

This Unit explained the intricacies of portfolio management, beginning with the fundamental concept of diversification. Diversification involves spreading investments across various assets and asset classes to mitigate risk and optimize returns, offering a well-balanced approach to wealth management.

Portfolio management is the central focus of this Unit. It means overseeing and optimizing investment collections, including activities such as asset allocation, risk assessment, and performance monitoring. The primary objective is to align portfolios with investors' financial goals, risk tolerance, and timeframes, achieved through thorough analysis and proactive decision-making. The stages of portfolio management include understanding investor preferences, formulating investment policies, selecting assets, tracking performance, and making necessary adjustments. This dynamic process necessitates constant attention and adaptation to changing market conditions. The Unit also explored the relationship between risk and return, essential for investment decision-making. Investors seek to strike a balance between risk and return, with strategies ranging from active management for potentially higher returns to passive management for market-based returns and reduced risk exposure. Risks inherent in investments include diversifiable and systematic risks, each requiring distinct risk management strategies. Diversification plays a crucial role in mitigating diversifiable risks, while

systematic risks impact the overall market and require broader risk management approaches. Furthermore, the Unit explained portfolio valuation and expected returns, crucial aspects influencing investment decisions. Portfolio valuation involves assessing the value of individual assets within a portfolio, while expected returns forecast potential profits or losses, guiding portfolio selection. In this Unit, we also explored Markowitz's Portfolio Theory, a seminal contribution to portfolio management. This theory emphasizes optimizing returns while managing risk through efficient portfolio construction. Despite its merits, Markowitz's theory faces criticisms, including unrealistic assumptions and challenges in implementation.

18.7 KEY WORDS

Diversification	: The practice of spreading investments across various assets and asset classes to mitigate risk and optimize returns.
Markowitz Portfolio theory	: An investment theory based on the idea that risk-averse investors can construct portfolios to optimize or maximize expected return based on a given level of market risk, emphasizing that risk is an inherent part of higher reward
Efficient Portfolio Frontier	: A set of investment portfolios that are expected to provide the highest returns at a given level of risk.
Portfolio	: A collection of investments owned by an individual or entity, which can include various types of assets like stocks, bonds, mutual funds, and real estate.
Portfolio management	: The practice of overseeing and optimizing portfolios. It involves carefully deciding how to allocate resources, balancing risk and returns, and adapting to changes in the market.
Diversifiable risk	: It refers to firm-specific risks that affect individual stock prices rather than the entire industry or sector.
Non-Diversifiable Risk	: The risks that affect the overall market or a specific sector. It is influenced by factors that impact the entire economy.
Expected Returns	: The anticipated profit or loss an investor can reasonably foresee from an investment within a specific timeframe

18.8 SOME USEFUL BOOKS

- Strong, R. A. (2006). Portfolio construction, management and protection. Thomson/South-Western.
- Mishkin, F. S., & Eakins, S. G. (2006). Financial markets and institutions. Pearson Education India.
- Counsel, I. (2008). Portfolio Management: Theory & Practice.
- Dicksler, James L. and Samuelson, Paul A., Investment Portfolio Decision-Making, London, Lexington Books, 1974.

18.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) A portfolio is essentially a collection of investments owned by an individual or entity, which can include various types of assets like stocks, bonds, mutual funds, and real estate. Stages: Understanding the goals, Creating and executing strategies, choosing asset classes and specific securities, assessing and tracking the portfolio's performance and analyzing the portfolio's outcomes
- 2) Consistent Risk Measurement, guidance for Strategic Planning, increased Transparency in Decision Making, understanding value creation, identifying improvement areas.

Check Your Progress 2

- 1) Refer to Sub-sections 18.3.2 and 18.3.3
- 2) Diversifiable risk, also known as unsystematic risk (unique risks), refers to firm-specific risks that affect individual stock prices rather than the entire industry or sector. Systematic risks, or non-diversifiable risks, are the risks that affect the overall market or a specific sector.

Check Your Progress 3

- 1) Risk-averse investors can construct portfolios to optimize or maximize expected return based on a given level of market risk. There is a direct relationship between the variability of returns and risk: the higher the variability, the higher the risk. Return can be maximised through a combination of securities whose total variability is lower.
- 2) An efficient portfolio is one where no higher expected return can be attained without accepting a higher level of risk. Conversely, it's a

portfolio where no greater certainty of returns can be obtained without sacrificing expected returns.

- 3) Unrealistic assumptions, Complexity and Practicality, Sensitivity to input data, neglect of transaction costs and taxes, single-period analysis, ignorance of Behavioural Finance insights, inability to capture tail risks.



UNIT 19 CAPITAL ASSET PRICING MODEL

Structure

- 19.0 Objectives
- 19.1 Introduction
- 19.2 The Capital Asset Pricing Model (CAPM)
 - 19.2.1 Distinctive Features of Capital Asset Pricing Model
- 19.3 Importance of Sharpe's Theory
 - 19.3.1 Objectives and Functions of Sharpe's Theory
 - 19.3.2 Importance and Role of Sharpe's Theory in an Economy
- 19.4 Application of Capital Asset Pricing Model
- 19.5 Limitation of CAPM
 - 19.5.1 Theoretical Limitation of CAPM
 - 19.5.2 Practical Limitation of CAPM
- 19.6 Empirical Analysis of the CAPM Model
- 19.7 Let Us Sum Up
- 19.8 Key Words
- 19.9 Some Useful Books
- 19.10 Answers/ Hints to Check Your Progress Exercises
- 19.11 Terminal Questions

19.0 OBJECTIVES

After completing this unit, you will be able to:

- explain the concept of CAPM;
- spell out the composition of Sharpe's theory;
- highlight the importance, role and functions of the CAPM model in an economy;
- identify assumptions of the CAPM model;
- describe the CAPM model;
- analyse the applications of the CAPM model and its empirical analysis; and
- indicate the problems of the CAPM model.

19.1 INTRODUCTION

You must have understood that financial markets play a very important role in the economy by facilitating the allocation of capital managing risk and enabling price discovery. They are influenced by various factors such as economic conditions, government policies, sentiment of investors and geopolitical events. A proper understanding and analysis of these markets are necessary for investors, policymakers and businesses to make informed decisions regarding investment, risk management and capital allocation. We have already discussed about Efficient Portfolio frontier and Markowitz's Portfolio theory in Unit 18. We have also discussed the relationship between risk and return in Unit 18. We will extend our understanding of risk, return and variance in this unit and discuss the Capital Asset Pricing Model (CAPM).

Models like CAPM provide valuable insights into the pricing and behaviour of these financial markets. CAPM provides valuable insight into the relationship between risk and return and financial markets. This unit will explore the foundational principles of CAPM and will also help in examining the implications for the economy as a whole. This unit will also help the learners in analyzing the role of CAPM in asset pricing, portfolio management and decision-making. This unit will also throw light on how CAPM influences economic outcomes and market efficiency. The limitations and critiques of CAPM will also be discussed.

19.2 THE CAPITAL ASSET PRICING MODEL (CAPM)

The Capital Asset Pricing Model (CAPM) is a fundamental framework in finance that plays a crucial role in asset pricing, portfolio management, and investment decision-making. Developed in the 1960s by William Sharpe, John Linter, and Jan Mossin, CAPM provides a theoretical foundation for understanding the relationship between risk and return in financial markets. Its significance in finance stems from its ability to quantify the expected return on an investment based on its level of systematic risk, offering valuable insights for investors, analysts, and policymakers.

Before delving into the key aspects of the CAPM model, let us discuss some key concepts that form the basis of CAPM. These are as follows:

Systematic Risk: We have read about systematic and non-systematic risk in Unit 18. Systematic risk, also known as market risk, refers to the risk that cannot be diversified away by holding a diversified portfolio. It is associated with factors that affect the entire market, such as economic downturns, interest rate changes, or geopolitical events.

Beta: Beta measures the sensitivity of an asset's returns to changes in the overall market returns. It quantifies the extent to which an asset moves in tandem with the market. Assets with a beta of 1 move in perfect correlation

with the market, while assets with a beta greater than 1 are more volatile, and those with a beta less than 1 are less volatile.

Risk-Free Rate: The risk-free rate represents the return on an investment with zero risk, typically provided by the yield on government bonds. It serves as the baseline return that investors can earn without taking on any investment risk.

Market Risk Premium: The market risk premium is the additional return that investors demand for bearing systematic risk. It is calculated as the difference between the expected return on the market portfolio and the risk-free rate.

Formulation of the CAPM Equation: The CAPM equation expresses the relationship between expected return, systematic risk (beta), and the risk-free rate:

$$\text{Expected Return} = R_f + \beta \times (R_m - R_f)$$

Where:

- Expected Return: Expected Return is the expected return on the asset.
- R_f is the risk-free rate, representing the return on investment with zero risk, typically proxied by the yield on government bonds.
- β is the asset's beta, which measures its systematic risk or sensitivity to market movements.
- R_m is the expected return on the market portfolio, usually represented by a broad market index like the S&P 500.
- $(R_m - R_f)$ is the market risk premium, representing the excess return investors demand for bearing the systematic risk of the market portfolio.

19.2.1 The Distinctive Features of the Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) is an extremely valuable tool to analyse the financial market. Some of the distinctive features of the model are as follows:

- 1) **Systematic Risk Emphasis:** CAPM focuses on systematic risk, also known as non-diversifiable risk, which is the risk inherent to the entire market or market segment. It assumes that investors are primarily concerned with this type of risk because it cannot be eliminated through diversification.
- 2) **Single-Factor Model:** CAPM is a single-factor model, relying on beta as the sole measure of risk. Beta measures an asset's sensitivity to market movements, providing a straightforward way to assess its risk relative to

the market portfolio. This simplicity makes CAPM easy to understand and apply.

- 3) **Relationship between Risk and Return:** CAPM quantifies the relationship between an asset's risk and its expected return. It asserts that investors require higher returns for bearing higher levels of systematic risk, as reflected in the market risk premium.
- 4) **Market Portfolio Benchmark:** CAPM uses the market portfolio as a benchmark for risk and return. The market portfolio represents a diversified portfolio containing all risky assets in the market, weighted according to their market capitalization. By comparing an asset's expected return to the return on the market portfolio, investors can assess its performance and risk-adjusted return.
- 5) **Risk-Free Rate Incorporation:** CAPM incorporates the risk-free rate, representing the return on an investment with zero risk. This element allows investors to differentiate between the risk-free return and the additional return required for bearing systematic risk, facilitating investment decision-making.
- 6) **Equilibrium Pricing:** CAPM is based on the concept of market equilibrium, suggesting that asset prices adjust to reflect the relationship between risk and return implied by the model. In equilibrium, the expected returns of all assets align with their systematic risk as measured by beta.
- 7) **Universal Applicability:** CAPM's principles apply across various asset classes and markets, making it a widely used model for determining expected returns on stocks, bonds, and other financial instruments. Its broad applicability enhances its utility in investment analysis and portfolio management.

19.3 IMPORTANCE OF SHARPE'S THEORY

Sharpe ratio is one of the indexes derived from the CAPM model. It is used to evaluate the value of investment in a portfolio. This section discusses the objectives, functions and application of the Sharpe ratio.

19.3.1 Objectives and Functions of Sharpe's Theory

The Sharpe ratio is a measure of risk-adjusted return, commonly used in finance to assess the performance of an investment or portfolio relative to its risk. The formula for the Sharpe ratio is:

$$\text{Sharpe ratio} = (R_p - R_f) / \sigma_p$$

Where:

- R_p is the expected return on the investment or portfolio.

- R_f is the risk-free rate of return (typically the yield on government bonds).
- σ_p is the standard deviation of the investment or portfolio's returns, representing its volatility or risk.
- The numerator ($R_p - R_f$) calculates the excess return of the investment or portfolio over the risk-free rate, while the denominator σ_p measures its volatility or risk. Dividing the excess return by the risk provides a measure of risk-adjusted return, with a higher Sharpe ratio indicating better risk-adjusted performance.

Its primary objective and functions in the economy include:

- Objective Measurement of Risk-Adjusted Performance:** The main objective of the Sharpe ratio is to provide a standardized measure of risk-adjusted performance. It evaluates the excess return of an investment or portfolio relative to its risk, typically measured as the standard deviation of returns. By considering both return and risk, the Sharpe ratio allows investors to compare investments consistently.
- Quantifying Risk-Return Tradeoff:** The Sharpe ratio quantifies the tradeoff between risk and return by expressing the excess return per unit of risk taken. A higher Sharpe ratio indicates better risk-adjusted performance, as it reflects higher returns relative to the volatility of those returns. This function helps investors evaluate whether an investment's return compensates adequately for the level of risk it entails.
- Portfolio Allocation Decision:** In portfolio management, the Sharpe ratio assists investors in making allocation decisions by comparing the risk-adjusted returns of different asset classes or investment strategies. It enables investors to identify portfolios or assets that offer the highest return per unit of risk and optimize their portfolio allocations accordingly.
- Performance Evaluation Tool:** The Sharpe ratio serves as a performance evaluation tool for investment managers, financial analysts, and fund selectors. It allows them to assess the skill of portfolio managers or the performance of investment funds relative to their benchmark or peer group. A higher Sharpe ratio suggests superior risk-adjusted performance, while a lower ratio may indicate underperformance or excessive risk-taking.
- Risk Management Tool:** The Sharpe ratio facilitates risk management by providing insights into the risk-adjusted returns of investment portfolios. Investors can use it to identify and manage portfolio risk exposures, ensure diversification, and optimize risk-return tradeoffs according to their risk preferences and investment objectives.

- vi) **Benchmark Comparison:** The Sharpe ratio enables investors to compare the risk-adjusted performance of their portfolios or investments against a benchmark, such as a market index or a target return. This comparative analysis helps investors gauge whether their portfolio's returns are commensurate with the level of risk taken relative to a chosen benchmark.

Thus, the Sharpe ratio plays a crucial role in the economy by providing a standardized measure of risk-adjusted performance, aiding portfolio allocation decisions, evaluating investment performance, managing portfolio risk, and facilitating benchmark comparisons. Its objective function is to help investors make informed decisions by assessing the risk-return tradeoff of investments or portfolios.

19.3.2 Importance and Role of Sharpe's Theory in an Economy

While the Sharpe ratio is more commonly used in the realm of finance and investment, its principles can still be applied to macroeconomic analysis, particularly in evaluating the risk-adjusted performance of economic policies or strategies. Let's consider some hypothetical examples to illustrate the importance and role of the Sharpe ratio in macroeconomics:

- i) **Evaluation of Fiscal Policy:** Suppose a government is considering implementing an expansionary fiscal policy to stimulate economic growth. They plan to increase government spending and reduce taxes to boost aggregate demand. However, this policy entails higher government debt and potential inflationary pressures. The Sharpe ratio can be used to assess the risk-adjusted effectiveness of this fiscal policy. We can calculate the ratio by comparing the expected increase in GDP (or other relevant macroeconomic indicators) to the associated increase in government debt or inflation volatility. If the Sharpe ratio for the proposed fiscal policy is high, it indicates that the potential benefits (e.g., higher GDP growth) outweigh the associated risks (e.g., increased debt or inflation). A low Sharpe ratio, on the other hand, suggests that the policy's risks may outweigh the benefits.
- ii) **Analysis of Monetary Policy:** Consider a Central Bank implementing an accommodative monetary policy by lowering interest rates to stimulate investment and consumption. While this policy may spur economic activity, it also carries risks such as asset price bubbles or currency depreciation. The Sharpe ratio can help evaluate the risk-adjusted impact of monetary policy actions. By comparing the expected increase in economic output or employment to the volatility of asset prices or exchange rates, policymakers can gauge the effectiveness and prudence of their monetary policy decisions. A high Sharpe ratio would indicate that the benefits of the monetary policy (e.g., higher economic growth) outweigh the risks (e.g., asset bubbles). Conversely, a low

Sharpe ratio suggests that the policy's risks may outweigh the benefits, signalling the need for alternative policy measures.

- iii) **Assessment of Structural Reforms:** Imagine a country implementing structural reforms aimed at improving long-term economic growth, such as deregulation, labour market reforms, or investment in infrastructure. While these reforms may yield significant economic benefits over time, they can also entail short-term disruptions or adjustment costs. The Sharpe ratio can be employed to assess the risk-adjusted effectiveness of structural reforms. By comparing the expected long-term benefits of the reforms to the short-term costs or uncertainties, policymakers can determine whether the reforms offer a favourable risk-return tradeoff. A high Sharpe ratio would suggest that the potential long-term benefits of the reforms justify the short-term costs or uncertainties. Conversely, a low Sharpe ratio may indicate that the risks associated with the reforms outweigh the expected benefits, prompting policymakers to reconsider their approach.

It is also important to note that while the application of the Sharpe ratio in macroeconomics may require adaptations and interpretations compared to its use in finance, it can still offer valuable insights into the risk-adjusted performance of economic policies and strategies. By quantifying the tradeoff between expected benefits and associated risks, policymakers can make more informed decisions to promote macroeconomic stability and sustainable growth.

19.4 APPLICATION OF CAPITAL ASSET PRICING MODEL

In financial economics, the Capital Asset Pricing Model (CAPM) serves as a fundamental tool for understanding the relationship between risk and return in financial markets. Here's how it's applied in financial economics, explained with an example:

Expected Return Estimation: CAPM is used to estimate the expected return of an asset or a portfolio based on its systematic risk, as measured by its beta (β). The formula for the expected return of an asset according to CAPM is:

$$E(R_i) = R_f + \beta_i \times (E(R_m) - R_f)$$

- Where:
- $E(R_i)$ is the expected return of asset i ,
- R_f is the risk-free rate,
- β_i is the beta of asset i ,
- $E(R_m)$ is the expected return of the market portfolio.

An example of this application could be estimating the expected return of a particular stock. Suppose the risk-free rate is 3%, the expected return of the market portfolio is 8%, and the stock's beta is 1.2. Using CAPM, we can calculate the expected return of the stock as follows:

$$\begin{aligned} &=0.03+1.2\times(0.08-0.03)=0.03+1.2\times0.05=0.03+0.06=0.09 \\ &=0.03+1.2\times(0.08-0.03)=0.03+1.2\times0.05=0.03+0.06=0.09 \end{aligned}$$

So, the expected return of the stock according to CAPM would be 9%.

Cost of Equity Calculation: CAPM is used to determine the cost of equity for a company, which represents the return required by investors for holding the company's stock. The cost of equity is a crucial input in various financial decisions, such as capital budgeting and valuation. Companies use CAPM to calculate their cost of equity by substituting the company's beta into the CAPM formula. For example, if a company has a beta of 1.5, the risk-free rate is 4%, and the market risk premium is 6%, then the company's cost of equity according to CAPM would be:

$$\begin{aligned} &=0.04+1.5\times0.06=0.04+0.09=0.13 \\ &0.13 \end{aligned}$$

So, the cost of equity for the company would be 13%.

Risk-Adjusted Performance Measurement: CAPM is used in financial economics to evaluate the performance of investment portfolios or funds. By comparing the actual returns of a portfolio with the returns predicted by CAPM based on the portfolio's risk (beta), investors can assess whether the portfolio manager has generated excess returns (alpha) or underperformed the market. For instance, if a portfolio's actual return is 10% and its expected return based on CAPM is 8%, the portfolio manager has generated an excess return of 2%, indicating positive alpha.

These are just a few examples of how CAPM is applied in financial economics to make informed decisions regarding asset pricing, cost of capital, and performance evaluation in financial markets.

Check Your Progress 1

- 1) Define Capital Asset Pricing Model.

.....

.....

.....

.....

- 2) Describe the main features of the Capital Asset Pricing Model.

.....

.....

-
-
- 3) Discuss the importance and role of the Capital Pricing Model in an economy.

.....

.....

.....

.....

- 4) List some applications of the Capital Pricing Model.

.....

.....

.....

.....

19.5 LIMITATION OF CAPITAL ASSET PRICING MODEL

Capital Asset Pricing Model (CAPM) is widely used and provides valuable insights into the relationship between risk and return in financial markets, it also has several limitations. Some of the key limitations include:

- 1) **Assumptions:** CAPM is based on several simplifying assumptions that may not always hold in real-world financial markets. These assumptions include perfect competition, frictionless markets, constant correlations among assets, and rational investor behaviour. Deviations from these assumptions can lead to inaccuracies in CAPM's predictions.
- 2) **Market Index:** CAPM relies on the market index to represent the overall market portfolio. However, the choice of the market index can significantly impact the results. Different market indices may have different compositions and characteristics, leading to variations in beta estimates and expected returns.
- 3) **Single-Factor Model:** CAPM is a single-factor model that considers only systematic risk (beta) as the sole determinant of expected returns. It does not account for other factors such as firm-specific risk, macroeconomic variables, or market sentiment, which can also influence asset prices and returns. As a result, CAPM may not fully capture the complexities of real-world asset pricing.
- 4) **Linearity and Homogeneity:** CAPM assumes linear and homogeneous relationships between risk and return, implying that all investors have the same risk preferences and risk aversion coefficients. In reality, investors

have diverse risk preferences and may exhibit nonlinear behaviour, leading to deviations from CAPM predictions.

- 5) **Risk-Free Rate:** CAPM relies on the risk-free rate as a proxy for the time value of money and the opportunity cost of capital. However, the risk-free rate may fluctuate over time and may not accurately reflect investors' actual borrowing and lending opportunities. Changes in the risk-free rate can impact the entire risk-return relationship predicted by CAPM.
- 6) **Empirical Validity:** Empirical studies have found mixed evidence regarding the validity of CAPM in explaining asset returns. While CAPM provides a useful benchmark for asset pricing, its ability to accurately predict returns in practice has been questioned, especially during periods of market turbulence or financial crises.
- 7) **Non-Market Risks:** CAPM focuses primarily on market (systematic) risk and may overlook non-market (idiosyncratic) risks that are specific to individual assets or industries. These non-market risks, such as company-specific events or regulatory changes, can significantly impact asset prices and returns but are not captured by CAPM.

Overall, while CAPM remains a foundational model in financial economics, it is essential to recognize its limitations and consider alternative models or factors when making investment decisions and assessing asset pricing in real-world financial markets.

19.5.1 Theoretical Limitation of Capital Asset Pricing Model

The theoretical limitation of the Capital Asset Pricing Model (CAPM) lies in its assumptions, which may not always hold in real-world financial markets. Some of the limitations are discussed as follows:

- 1) **Perfect Market Assumptions:** CAPM assumes the existence of perfect markets, where investors have access to all relevant information, transaction costs are zero, and there are no restrictions on borrowing or lending. In reality, markets are often imperfect, with information asymmetries, transaction costs, and regulatory constraints affecting investor behaviour and asset prices. For example: During the global financial crisis of 2007-2008, financial markets experienced significant disruptions, with widespread information asymmetries, liquidity shortages, and regulatory interventions. These market imperfections challenged the assumptions of perfect markets underlying CAPM, leading to distortions in asset pricing and deviations from CAPM predictions.
- 2) **Homogeneous Investor Behaviour:** CAPM assumes that all investors have homogeneous expectations, risk preferences, and investment horizons. In reality, investors exhibit diverse behaviour, with varying

risk aversion levels, investment goals, and time horizons. This heterogeneity in investor behaviour can lead to discrepancies between CAPM-predicted asset prices and actual market prices. For Example: During periods of market volatility, investors may exhibit different risk preferences and reactions to market events. Some investors may adopt a more risk-averse stance and demand higher returns for bearing risk, while others may be more willing to take on risk in pursuit of higher returns. These heterogeneous behaviours can lead to deviations from the CAPM-predicted relationship between risk and return.

- 3) **Single-Factor Model:** CAPM is a single-factor model that considers only systematic risk (beta) as the sole determinant of expected returns. It does not account for other factors such as firm-specific risk, macroeconomic variables, or market sentiment, which can also influence asset prices and returns. This limitation restricts CAPM's ability to fully capture the complexities of asset pricing dynamics. For Example, The financial markets often experience periods of heightened uncertainty driven by macroeconomic events, geopolitical tensions, or changes in investor sentiment. During such periods, factors beyond systematic risk, such as changes in interest rates, inflation expectations, or regulatory policies, can exert significant influence on asset prices, leading to deviations from CAPM predictions.

These theoretical limitations highlight the challenges in applying CAPM in real-world financial markets and underscore the importance of considering alternative models or factors to better understand asset pricing dynamics and make informed investment decisions.

19.5.2 Practical Limitation of Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) is a valuable tool in finance, but it does have practical limitations.

- 1) **Non-Linear Relationship between Risk and Return:** CAPM assumes a linear relationship between risk (measured by beta) and expected return. However, in reality, the relationship between risk and return may not always be linear. Assets with higher betas may not always provide proportionally higher returns, and the relationship may exhibit non-linearity, especially at extreme ends of the risk spectrum. This limitation can be explained with the help of an example of a Tech Start-Up. Consider the case of investing in technology startups. These startups often exhibit high levels of risk due to factors such as uncertain revenue streams, competition, and technological disruption. According to CAPM, investors would expect higher returns for investing in these high-risk ventures. However, the relationship between risk and return may not be strictly linear. While some successful startups may provide significant returns, many others may fail, leading to losses for investors despite their

high-risk nature. This non-linear relationship challenges CAPM's ability to accurately predict returns for such high-risk investments.

- 2) **Sensitivity to the inputs and assumptions used in its calculations** can lead to unreliable estimates of expected returns. CAPM relies on several inputs and assumptions, including the risk-free rate, market risk premium, and beta coefficients. However, these inputs are subject to uncertainty and may vary over time, leading to potential inaccuracies in CAPM's predictions. Small changes in these inputs can result in significant fluctuations in estimated expected returns, reducing the reliability of CAPM as a forecasting tool. This limitation is also explained with the help of the following example of Sensitivity to market risk and premium. Consider the estimation of the market risk premium, which represents the excess return investors expect from investing in the market portfolio compared to the risk-free rate. This parameter is crucial in determining expected returns according to CAPM. However, the market risk premium is not directly observable and must be estimated based on historical data or other methodologies. If different analysts or investors use different methods or data sources to estimate the market risk premium, they may arrive at divergent estimates. Additionally, the market risk premium itself may fluctuate over time due to changes in economic conditions, investor sentiment, or market dynamics. As a result, small variations in the estimated market risk premium can lead to significant differences in expected returns according to CAPM. The sensitivity of CAPM to its inputs and assumptions can have practical implications for investment decision-making. Investors and financial analysts may struggle to obtain accurate estimates of expected returns using CAPM, especially in dynamic or uncertain market environments. Relying solely on CAPM's predictions without considering the inherent uncertainty and sensitivity of its inputs can result in suboptimal investment decisions and portfolio performance.
- 3) **Poor predictor of returns:** Empirical studies support that the reliability of CAPM becomes weak when the estimate of asset returns is closely associated with realized returns. Empirical findings do not support the hypothesis that asset returns are only determined by systematic risk. The low predictive power of the CAPM model is one of the serious limitations because investors invest based on a prediction of future returns.

Check Your Progress 2

- 1) Discuss the limitations of the Capital Asset Pricing Model.

.....

.....

.....

2) Describe the following:

(a) Theoretical Limitation of Capital Asset Pricing Model.

.....
.....
.....

(b) A practical limitation of the Capital Asset Pricing Model.

.....
.....
.....

19.6 EMPIRICAL ANALYSIS OF THE CAPM MODEL

The Capital Asset Pricing Model (CAPM) can be applied to real-world financial data and the results can be tested with its predictions. Some of the applications of the CAPM model are as follows:

Time-Series Analysis: Economists collect historical data on individual assets' returns and the overall market return over a specified period. Using regression analysis, they estimate the beta coefficients for each asset to measure their sensitivity to market movements. CAPM's predictions are then tested by comparing the expected returns calculated using CAPM with the actual historical returns of the assets. Example: A study examines the relationship between the beta coefficients of different stocks and their subsequent returns over 10 years. Economists use CAPM to estimate the expected returns and assess whether they closely match the actual returns observed in the market.

Cross-Sectional Analysis: Economists examine the relationship between asset returns and their beta coefficients across different assets at a specific point in time. Using statistical tests, they assess whether there is a significant positive relationship between beta and expected returns, as predicted by CAPM. An example: A study analyzes a sample of stocks from different industries and calculates the beta coefficients for each stock. Researchers then compare the expected returns derived from CAPM with the actual returns of the stocks to determine if CAPM holds across a diverse set of assets.

Event Studies: Researchers conduct event studies to examine the impact of specific events, such as earnings announcements or mergers, on asset returns. They use CAPM to predict how asset prices should respond to these events based on their beta coefficients and market returns.

Example: A study investigates the stock market's reaction to quarterly earnings announcements. Researchers calculate the expected returns using

CAPM and compare them with the actual returns observed around the announcement dates to evaluate CAPM's ability to explain asset price movements.

Portfolio Performance Evaluation: Investors use CAPM to assess the performance of their investment portfolios relative to their level of systematic risk. They calculate the portfolio's beta and compare its expected return, as predicted by CAPM, with the actual return achieved by the portfolio. Example: An investor analyzes the performance of their diversified portfolio of stocks over the past year. They use CAPM to estimate the expected return of the portfolio based on its beta and compare it with the actual return to evaluate whether the portfolio outperformed or underperformed relative to its level of risk.

Let us illustrate how empirical analysis of CAPM can be conducted using different methodologies and datasets to assess the model's validity and applicability in real-world financial markets. Suppose in a market:

- Risk-free rate (R_f): 3%
- Market return ($E(R_m)$): 8%
- Stock A's beta (β_A): 1.2

Using CAPM, we can calculate the expected return ($E(R_A)$) for Stock A as follows:

$$E(R_A) = R_f + \beta_A \times (E(R_m) - R_f)$$

Substituting the given values:

$$= 0.03 + 1.2 \times (0.08 - 0.03) \quad E(R_A) = 0.03 + 1.2 \times (0.08 - 0.03)$$

$$= 0.03 + 1.2 \times 0.05 \quad E(R_A) = 0.03 + 1.2 \times 0.05$$

$$= 0.03 + 0.06 \quad E(R_A) = 0.03 + 0.06$$

$$= 0.09 \quad E(R_A) = 0.09$$

So, according to CAPM, the expected return for Stock A is 9%.

Now, let's assume that the actual return for Stock A over a specific period was 10%. We can then assess whether this actual return aligns with the expected return predicted by CAPM.

We can compare the actual return with the expected return:

In this case, the Actual return i.e., 10% is greater than the Expected return i.e., 9%. Since the actual return for Stock A is greater than the expected return calculated using CAPM, this suggests that Stock A outperformed relative to its level of systematic risk, as predicted by CAPM.

19.7 LET US SUM UP

CAPM is a fundamental model in finance that establishes a relationship between risk and return for individual assets or portfolios in a well-diversified market.

CAPM focuses on systematic risk, which cannot be diversified away, and measures it using beta (β), representing the asset's sensitivity to market movements.

According to CAPM, the expected return of an asset is determined by the risk-free rate (R_f) plus a risk premium proportional to the asset's beta and the market risk premium ($E(R_m) - R_f$).

The formula for expected return $E(R)$ using CAPM is: $E(R) = R_f + \beta \times (E(R_m) - R_f)$

A risk-free rate represents the return on a risk-free investment, typically proxied by government bonds, with no default risk.

The excess return investors demand for holding a risky asset compared to a risk-free asset. It is calculated as the difference between the expected return of the market portfolio $E(R_m)$ and the risk-free rate (R_f).

A theoretical portfolio that contains all assets in the market, weighted by their market values. It represents the average level of risk in the market.

CAPM implies that all investors will hold a combination of the risk-free asset and the market portfolio, known as the efficient frontier, to achieve optimal risk-return trade-offs. It relies on several assumptions, including perfect capital markets, rational investors, homogeneous expectations, and constant risk aversion. CAPM is widely used in finance for asset pricing, portfolio management, and performance evaluation. It provides a benchmark for determining whether an asset's expected return adequately compensates for its level of risk.

Despite its widespread use, CAPM has limitations, including its simplifying assumptions, reliance on historical data, inability to capture non-market risks, and mixed empirical evidence.

Researchers have developed alternative models and extensions to CAPM, such as multi-factor models and behavioural finance frameworks, to address its limitations and improve the understanding of asset pricing dynamics.

To conclude, CAPM serves as a foundational framework for understanding the relationship between risk and return in financial markets, guiding investment decisions and portfolio management strategies. However, it is essential to recognize its assumptions and limitations when applying them in real-world contexts.

19.8 KEY WORDS

Anomalies	: Patterns or inconsistencies in asset returns that are not explained by CAPM.
Asset Pricing	: The process of valuing assets based on their risk-return characteristics.
Beta	: The measure of an asset's sensitivity to market movements.
Diversification	: Spreading investments across different assets to reduce risk.
Efficient Frontier	: The set of optimal portfolios that offer the highest expected return for a given level of risk.
Empirical Analysis	: Examination of real-world data to test the validity and applicability of economic theories like CAPM.
Expected Return	: The return an investor anticipates receiving on an investment is based on its risk characteristics.
Market Efficiency	: The degree to which prices of assets accurately reflect all available information in the market.
Market Portfolio	: A theoretical portfolio containing all assets in the market, weighted by their market values.
Market Risk Premium	: Excess return expected from holding a risky asset compared to risk-free asset.
Portfolio Theory	: The study of how investors can construct portfolios to optimize returns while minimizing risk.
Return	: The gain or loss generated on an investment over a specific period.
Risk	: The variability of returns associated with an investment.
Risk-Free Rate	: The return on a risk-free investment, is usually proxied by government bonds.
Systematic Risk	: Market-related risks that cannot be diversified away

19.9 SOME USEFUL BOOKS

Abel B (1990) "Asset Prices under Habit Formation and catching up with the Jones; The American Review 80(2) pp 38-42

Chen N Roll and Ross S (1986) “Economic forces and the Stock Market
Journal of Business 59(3) pp 383-403

Frank J Faboozi (2006) “Financial Modeling of the Equity Market from
CAPM to Cointegration.

Haim Levy (2011) The Capital Asset Pricing Model in the 21st Century

Martin L and Gonzalo R (2011) Evaluating alternative methods forecasting
asset pricing models with historical data. “Journal of Empirical Finance 18(1)
pp 136-146

19.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) CAPM provides a theoretical foundation for understanding the relationship between risk and return in financial markets.
- 2) Emphasis on systematic risk, single-factor model, relationship between risk and return, market portfolio benchmark, risk-free rate incorporation, equilibrium pricing, universal applicability.
- 3) Objective measurement of risk-adjusted performance, quantifying risk-return tradeoff, portfolio allocation decision, performance evaluation tool, risk management tool, benchmark comparison.
- 4) Expected return estimation, cost of equity calculation, risk-adjusted performance measurement.

Check Your Progress 2

- 1) Strict assumptions, market index, single-factor model, linearity and homogeneity, risk-free rate, empirical validity, non-market risks.
- 2) a) Perfect Market Assumptions, Homogeneous Investor Behaviour, Single-Factor Model.

b) Non-linear relationship between risk and return, sensitivity to the inputs and assumptions used in its calculations, poor predictor of returns.

19.11 TERMINAL QUESTIONS

- 1) What is the Capital Asset Pricing Model (CAPM), and what are its key assumptions?
- 2) Provide the CAPM equation and explain the significance of each component.

- 3) Discuss common statistical measures used to evaluate the goodness-of-fit of CAPM in empirical analyses.
- 4) Explain how the presence of anomalies affects the applicability of CAPM in real-world financial markets.
- 5) Discuss the theoretical and practical limitations of CAPM.



ignou
THE PEOPLE'S
UNIVERSITY

UNIT 20 ARBITRAGE PRICING THEORY

Structure

- 20.0 Objectives
- 20.1 Introduction
- 20.2 Ross's Critique of the Capital Asset Pricing Model (CAPM)
 - 20.2.1 Drawbacks of CAPM
 - 20.2.2 Simplifying Assumptions of CAPM
- 20.3 Introduction to Arbitrage Pricing Theory (APT)
 - 20.3.1 Assumptions of Arbitrage Pricing Theory (APT)
 - 20.3.2 Theoretical Foundations of APT
 - 20.3.3 Key Differences between APT and CAPM
- 20.4 Empirical Studies on APT
- 20.5 Criticism of the APT
 - 20.5.1 Critical Evaluation of APT's Assumptions
 - 20.5.2 Comparison of APT with Other Asset Pricing Models
 - 20.5.3 Identifying Areas of Improvement for APT
- 20.6 Let Us Sum Up
- 20.7 Key Words
- 20.8 Some Useful Books
- 20.9 Answers/ Hints to Check Your Progress Exercises

20.0 OBJECTIVES

After going through this unit, you will be able to:

- explain the limitations of the CAPM Model;
- describe Ross's critique and extension of the CAPM model;
- describe assumptions of the APT; and
- discuss its empirical study extensions.

20.1 INTRODUCTION

In the intricate realm of financial markets, theories and models serve as essential tools for unravelling the complexities of asset pricing. This Unit embarks on a journey to explore, critique, and extend key asset pricing models, focusing primarily on the Capital Asset Pricing Model (CAPM) and the Arbitrage Pricing Theory (APT).

As we know from Unit 19, the CAPM, born in the 1960s through the collaborative efforts of William Sharpe, John Lintner, and Jan Mossin, laid a foundational understanding of expected returns about systematic risk. However, as financial markets evolved, critiques arose, with Stephen A. Ross leading the charge. Ross's critique of the CAPM centred on its reliance on a single market factor to explain asset prices. In his influential work, "The Arbitrage Theory of Capital Asset Pricing," Ross proposed an alternative approach, suggesting that multiple risk factors impact asset prices. This critique prompted a paradigm shift, urging scholars to move beyond the simplicity of the CAPM towards more nuanced models that could better capture the intricacies of financial markets. We have already discussed about CAPM model in the previous unit, let us discuss its criticism and the APT model as an alternative to the CAPM model.

20.2 ROSS'S CRITIQUE OF THE CAPITAL ASSET PRICING MODEL (CAPM)

Stephen A. Ross, who developed the Arbitrage Pricing Theory (APT), pointed out several drawbacks and simplifying assumptions of the Capital Asset Pricing Model (CAPM). Let us briefly discuss the drawbacks and simplifying assumptions of CAPM model once again.

20.2.1 Drawbacks of CAPM

- 1) **Single-Factor Model:** CAPM is a single-factor model that relies heavily on the market risk factor (beta). It assumes that all systematic risk can be captured by a single factor, which might oversimplify the real-world complexities of asset pricing.
- 2) **Homogeneous Expectations:** CAPM assumes that all investors share the same expectations about future returns, standard deviations, and correlations. This assumption overlooks the diversity of investor beliefs and expectations in the market.
- 3) **Static Risk:** CAPM assumes that the risk associated with an asset is constant over time. In reality, the risk profile of an asset may change due to various factors such as economic conditions, technological advancements, or changes in market sentiment.
- 4) **Risk-Free Rate Assumption:** The model relies on a risk-free rate as a baseline. However, the actual risk-free rate might not be constant, and using government bond yields as a proxy for the risk-free rate assumes that there is no default risk, which may not always hold.

20.2.2 Simplifying Assumptions of CAPM

- 1) **Perfect Markets:** CAPM assumes the existence of perfect and frictionless financial markets. This implies that there are no transaction costs, no taxes, and no restrictions on short selling. In reality, markets are not always perfect, and these assumptions might not hold.

- 2) **Homogeneous Assets:** CAPM assumes that all assets are homogeneous and can be completely described by their expected return and beta. In reality, assets may have unique characteristics that are not captured by these simplified parameters.
- 3) **Normal Distribution of Returns:** The model assumes that asset returns are normally distributed. However, in practice, financial markets often exhibit non-normalities, such as fat tails and volatility clustering, which are not accounted for in the CAPM.
- 4) **Constant Correlations:** CAPM assumes constant correlations between assets. In reality, correlations can change over time, especially during periods of financial distress or market turbulence.
- 5) **Investor Rationality:** CAPM assumes that investors are rational and make decisions solely based on expected returns and risks. Behavioural finance research suggests that investors may not always behave rationally, and psychological factors can influence decision-making.

Understanding these drawbacks and simplifying assumptions is crucial for recognizing the limitations of CAPM and considering alternative models, such as the Arbitrage Pricing Theory (APT), which seeks to address some of these issues by allowing for multiple factors and a more flexible framework.

Check Your Progress 1

- 1) Discuss Ross's critique of CAPM, highlighting drawbacks.
.....
.....
.....
- 2) What were some simplifying assumptions of the CAPM model?
.....
.....
.....

20.3 INTRODUCTION TO ARBITRAGE PRICING THEORY

Arbitrage Pricing Theory (APT) is an asset pricing model that was developed by economist Stephen Ross in the early 1970s as an alternative to the Capital Asset Pricing Model (CAPM). The APT is designed to provide a more flexible framework for explaining the expected returns on assets by allowing for multiple sources of systematic risk. Unlike CAPM, APT doesn't rely on the assumption of a single systematic risk factor (market risk), and it does not specify the nature or number of risk factors in advance.

Stephen Ross introduced APT in his seminal paper titled “The Arbitrage Theory of Capital Asset Pricing” published in 1976. The theory was developed as a response to perceived limitations and simplifying assumptions of CAPM. APT acknowledges that an asset’s expected return is influenced by various macroeconomic factors, and it seeks to model these influences more realistically and dynamically.

20.3.1 Assumptions of Arbitrage Pricing Theory (APT)

Systematic and Unsystematic Risk in APT:

- 1) **Separation of Systematic and Unsystematic Risk:** A key assumption of APT is the separation of systematic and unsystematic risk. Systematic risk is captured by the sensitivity of an asset’s return to various macroeconomic factors (as represented by factor betas in the APT model). Unsystematic risk, also known as specific or idiosyncratic risk, is assumed to be diversifiable and can be eliminated through a well-diversified portfolio. APT assumes that investors can eliminate unsystematic risk through proper portfolio construction. We have discussed about portfolio construction in Unit 18.

Market Efficiency and Rational Expectations:

- 2) **Efficient Markets:** APT assumes that financial markets are efficient. In an efficient market, asset prices fully reflect all available information, and mispricing is quickly corrected through the actions of rational and profit-seeking investors. This assumption is crucial for the functioning of arbitrage opportunities, as any deviation from fair value would be exploited by arbitrageurs.
- 3) **Rational Expectations:** APT assumes that investors have rational expectations, meaning they form their expectations about future asset prices based on all available information. Rational expectations are essential for the correct pricing of assets in an efficient market and for the proper functioning of arbitrage.

Arbitrage Opportunities and Their Impact on Asset Pricing:

- 4) **Absence of Arbitrage Opportunities:** A fundamental assumption in APT is the absence of arbitrage opportunities. If arbitrage opportunities were present, investors could make risk-free profits by exploiting mispricing, leading to market adjustments and undermining the model’s ability to explain asset pricing. The assumption of no arbitrage ensures that asset prices accurately reflect their underlying risks.

Rational Behaviour of Investors in APT:

- 5) **Investor Rationality:** APT assumes that investors are rational decision-makers. Rational behaviour implies that investors make choices that maximize their expected utility, given their information and preferences.

Rationality is crucial for the proper functioning of arbitrage and for ensuring that asset prices align with the expectations implied by the model.

In summary, the assumptions of APT revolve around the separation of systematic and unsystematic risk, the efficiency of financial markets, the rational expectations of investors, and the absence of arbitrage opportunities. These assumptions provide the foundation for the model's ability to explain asset pricing dynamics based on the influence of multiple macroeconomic factors.

20.3.2 Theoretical Foundations of APT

The Arbitrage Pricing Theory (APT) provides a framework for understanding the relationship between risk and expected return in financial markets. It was developed by Stephen Ross in 1976 as an alternative to the Capital Asset Pricing Model (CAPM).

The APT formula for expected asset returns is given by:

$$E(R_i) = R_f + \beta_{i1} \times \lambda_1 + \beta_{i2} \times \lambda_2 + \dots + \beta_{ik} \times \lambda_k + \epsilon_i$$

Where:

- $E(R_i)$ is the expected return on asset i ,
- R_f is the risk-free rate,
- β_{ij} represents the sensitivity of asset i to factor j ,
- λ_j is the risk premium associated with factor j ,
- ϵ_i is the specific or idiosyncratic return component.

No Arbitrage Assumption:

APT operates under the assumption of no arbitrage opportunities in the market. This implies that investors cannot consistently earn risk-free profits by exploiting price discrepancies between assets. The absence of arbitrage opportunities ensures that asset prices are efficient and reflect all available information.

Law of One Price:

APT relies on the law of one price, which states that assets with similar risk exposures should have similar expected returns to prevent arbitrage opportunities. In other words, assets that share common risk factors should be priced in such a way that investors are indifferent between them.

Flexibility and Dynamism:

One of the key advantages of APT is its flexibility and dynamism in pricing assets. Unlike CAPM, which relies on a single systematic risk factor (market risk), APT accommodates multiple systematic risk factors. This allows for a

more nuanced understanding of asset pricing, as different assets may be influenced by different sets of risk factors.

Empirical Identification of Factors:

APT allows for the empirical identification of systematic risk factors based on historical data analysis. By analysing the relationship between asset returns and various macroeconomic, financial, and market variables, researchers can identify the factors that drive asset prices and determine their risk premiums.

Significance and Contribution:

The development of APT was a significant contribution to the field of finance, as it offered an alternative perspective to the traditional CAPM framework. By incorporating multiple systematic risk factors and allowing for empirical testing, APT provides a more comprehensive approach to asset pricing that better reflects the complexities of real-world markets.

In summary, APT provides a rigorous framework for understanding the relationship between risk and expected return in financial markets. Its reliance on the absence of arbitrage opportunities, the law of one price, and the accommodation of multiple risk factors make it a valuable tool for both academics and practitioners in the field of finance.

20.3.3 Key Differences between APT and CAPM

1) Number of Factors:

- **CAPM:** The Capital Asset Pricing Model (CAPM) relies on a single systematic risk factor, which is market risk. It assumes that all systematic risk can be captured by the market portfolio.
- **APT:** In contrast, the Arbitrage Pricing Theory (APT) allows for the consideration of multiple systematic risk factors. It recognizes that assets may be influenced by a variety of economic variables such as interest rates, inflation, GDP growth, and industry-specific factors.

2) Factor Identification:

- **CAPM:** CAPM assumes a specific market factor, represented by the market portfolio. This market factor is usually proxied by a broad-based equity index such as the S&P 500.
- **APT:** APT does not predefine factors. Instead, it allows researchers to empirically identify relevant factors based on statistical analysis of historical data. Factors may vary depending on the market and the specific assets being analysed.

3) Sensitivity Measures:

- **CAPM:** In CAPM, the sensitivity of an asset to systematic risk is measured by beta (β). Beta represents the covariance between the

asset's returns and the market portfolio returns, divided by the variance of the market portfolio returns.

- **APT:** APT uses factor sensitivities (β_{ij}). Each beta (β_{ij}) represents the i th asset's sensitivity to a specific factor (j). These factor sensitivities capture how changes in each factor affect the returns of the asset.

4) Assumption about Investor Behaviour:

- **CAPM:** CAPM assumes that investors are rational and risk-averse. It suggests that investors seek to maximize their utility by balancing risk and return and will invest in portfolios that offer the highest expected return for a given level of risk.
- **APT:** APT does not make explicit assumptions about investor behaviour. Instead, it focuses on the relationships between asset returns and macroeconomic factors. It allows for a more flexible and dynamic approach to asset pricing by accommodating different views on investor behaviour.

In summary, while both APT and CAPM are asset pricing models that relate expected returns to systematic risk, they differ in terms of the number of factors considered, the method of factor identification, the sensitivity measures used, and the assumptions about investor behaviour. APT's flexibility in accommodating multiple factors and empirical factor identification makes it a valuable tool for asset pricing analysis in real-world financial markets.

Check Your Progress 2

- 1) What are the key assumptions of the Arbitrage Pricing Theory (APT) regarding the separation of risk, market efficiency, and rational expectations?
.....
.....
.....
- 2) Provide the APT formula for expected asset returns and explain the significance of the assumptions related to the absence of arbitrage opportunities and investor rationality.
.....
.....
.....
- 3) Highlight one major difference between the Arbitrage Pricing Theory (APT) and the Capital Asset Pricing Model (CAPM) in terms of the number of systematic risk factors considered, and briefly discuss the implications of this difference.

.....
.....
.....

20.4 EMPIRICAL STUDIES ON APT

The Arbitrage Pricing Theory (APT) exhibits practical versatility in various applications, particularly in asset allocation, cost of capital computation, and the evaluation of managed funds. The utilization of APT in asset allocation stems from its connection to the factor structure and mean-variance efficiency. With k factors implying the existence of efficient frontier-spanning assets, constructing a mean-variance efficient portfolio becomes more straightforward when k factors are the payoffs of traded securities. APT simplifies the optimization problem by reducing dimensionality when k is small. However, if the factor structure is incorrect, the resulting portfolio may not be mean-variance efficient, necessitating adjustments based on investors' beliefs, as proposed by Pastor and Stambaugh (2000) using Bayesian methods.

In the context of calculating the cost of capital, APT has been applied by various researchers, such as Elton, Gruber, and Mei (1994) and Bower and Schink (1994). However, the challenge lies in the selection of factors, with different studies yielding disparate results. This lack of factor specification is a significant drawback, contributing to the reluctance of entities like the U.S. Federal Reserve Board to adopt APT for certain purposes.

APT's application in evaluating money managers involves regressing managed funds' returns on factors and comparing intercepts with benchmark securities returns. Notable studies, including those by Busse (1999) and Carhart (1997), have explored this application. However, it is essential to recognize the one-period nature of APT and its limitations when pricing derivatives or trading strategies due to potential arbitrage issues arising from the stochastic discount factor.

The APT's ironic challenge is evident in the need for extensions to the basic model in some applications, violating its core tenet of market arbitrage absence. For instance, evaluating dynamic trading strategies may require alternative models like Black-Scholes, as suggested by Glosten and Jagannathan (1994), highlighting the complexity and limitations in the practical application of APT.

Identification of Challenges and Criticisms from Empirical Studies:

- 1) **Data Limitations:** Some empirical studies face challenges related to data limitations. Identifying and collecting accurate historical data for various macroeconomic factors and asset returns can be complex, and the quality of results may depend on the availability and reliability of the data.

- 2) **Model Misspecification:** Critics argue that APT, like any model, is subject to misspecification. If the chosen factors do not adequately capture the relevant sources of systematic risk, the model may provide inaccurate predictions. Empirical studies have explored the sensitivity of APT to the selection of factors.
- 3) **Time-Varying Betas:** Empirical evidence suggests that factor sensitivities (betas) may not be constant over time, challenging the assumption of a static risk profile in APT. Researchers have explored the implications of time-varying betas on the model's performance.
- 4) **Factor Redundancy:** Some studies have found that identified factors may be correlated or redundant, raising questions about the independence of factors assumed by APT. Redundancy in factors could affect the reliability of the model's predictions.
- 5) **Market Efficiency Concerns:** Empirical studies testing APT often assume market efficiency, but real-world markets may deviate from perfect efficiency. If markets are not fully efficient, it could impact the ability of arbitrage to correct mispricing and, consequently, the validity of APT predictions.

In summary, empirical studies on APT have provided valuable insights into the model's performance in explaining asset pricing. While some findings support the theory, challenges and criticisms related to data limitations, model misspecification, time-varying betas, factor redundancy, and market efficiency concerns have been identified. Researchers continue to refine and extend empirical testing to enhance our understanding of APT's strengths and limitations in capturing the complexities of asset pricing.

20.5 CRITICISMS OF APT

20.5.1 Critical Evaluation of APT's Assumptions

- 1) **Arbitrage Opportunities:** Critics argue that the assumption of no arbitrage opportunities might be too restrictive. In real markets, various frictions, transaction costs, and constraints may limit the ability of arbitrageurs to correct mispricing quickly and efficiently.
- 2) **Market Efficiency:** APT relies on the assumption of efficient markets, where prices fully reflect all available information. In practice, markets may not always be perfectly efficient, leading to concerns about the model's applicability in less-than-perfectly-efficient environments.
- 3) **Factor Independence:** APT assumes that factors influencing asset returns are independent. Critics question the independence of these factors, suggesting that correlations or dependencies may exist, challenging the model's assumption of uncorrelated risk factors.
- 4) **Static Betas:** While APT allows for multiple factors, it often assumes that factor sensitivities (betas) are constant over time. Critics argue that

betas can be dynamic, changing with economic conditions and market dynamics and that the static nature of betas may limit the model's accuracy.

- 5) **Investor Rationality:** APT assumes rational expectations, but behavioural finance research suggests that investors may not always behave rationally. Behavioural biases and sentiment-driven behaviour may influence asset pricing, challenging the assumption of investor rationality.

20.5.2 Comparison of APT with Other Asset Pricing Models

- 1) **Comparison with CAPM:** APT has been compared with the Capital Asset Pricing Model (CAPM). Critics argue that while APT addresses some of the limitations of CAPM, it introduces its own set of assumptions and challenges. The choice between APT and CAPM often depends on the specific characteristics of the data and the underlying assumptions deemed most appropriate.
- 2) **Challenges Compared to the Fama-French Model:** The Fama-French Three-Factor Model, an extension of APT, includes market risk, size (SMB), and value (HML) factors. Some critics argue that the Fama-French model may offer better explanatory power than APT, especially in capturing size and value effects observed in empirical studies.

20.5.3 Identifying Areas of Improvement for APT

- 1) **Dynamic Factor Sensitivities:** Critics suggest that allowing for time-varying factor sensitivities in APT could enhance the model's ability to capture changing risk profiles over different market conditions.
- 2) **Incorporating Behavioural Aspects:** Integrating behavioural finance concepts into APT could address the limitations associated with the assumption of investor rationality. Considering psychological factors and market sentiment may provide a more realistic depiction of asset pricing dynamics.
- 3) **Enhancing Factor Identification:** Improving methods for identifying relevant factors is crucial. Empirical studies and model testing should focus on refining the process of selecting macroeconomic variables that genuinely drive asset returns, avoiding redundancy and ensuring independence.
- 4) **Accounting for Nonlinear Relationships:** APT typically assumes linear relationships between factors and asset returns. Acknowledging the possibility of nonlinear relationships may lead to a more accurate representation of the complexities in pricing dynamics.

In conclusion, while APT has been a valuable tool in asset pricing, it is not without criticisms. Addressing issues related to assumptions, considering alternatives, and exploring areas of improvement could contribute to the

refinement and evolution of the model in capturing the nuances of real-world financial markets.

Check Your Progress 3

- 1) What are some challenges and criticisms identified in empirical studies of the Arbitrage Pricing Theory (APT)?

.....
.....
.....

- 2) In the context of APT's comparison with other asset pricing models, what are the key considerations and criticisms?

.....
.....
.....

20.6 LET US SUM UP

The Arbitrage Pricing Theory (APT), introduced by Stephen Ross, serves as an alternative to the Capital Asset Pricing Model (CAPM) by incorporating multiple systematic risk factors influencing asset pricing. APT posits that the expected return on an asset is a linear function of various macroeconomic factors, with factor sensitivities (betas) determining the asset's response to each factor. Its applications extend to various financial markets, including equities, commodities, real assets, and fixed-income securities, with international considerations for cross-border asset pricing. Empirical studies have tested APT, identifying factors, assessing explanatory power, and addressing challenges like data limitations and model misspecification. Despite criticisms regarding assumptions and comparisons with other models, APT remains crucial for understanding asset pricing dynamics due to its flexibility and empirical testability. Ongoing research focuses on enhancing factor identification, incorporating dynamic models, considering behavioural aspects, and exploring non-linear relationships, highlighting APT's adaptability and potential for future advancements in the ever-evolving landscape of finance.

20.7 KEY WORDS

Arbitrage Pricing Theory (APT) : APT is an asset pricing model that offers an alternative to CAPM. It suggests that the expected return on an asset depends on multiple systematic risk factors, allowing for a more flexible framework compared to the single-factor CAPM.

Capital Asset Pricing Model (CAPM) : CAPM is a financial model that establishes a relationship between the expected return on an

investment and its systematic risk. It quantifies the expected return as a function of the risk-free rate, beta (systematic risk measure), and the market risk premium.

- Efficient Market Hypothesis (EMH)** : EMH asserts that financial markets efficiently incorporate all available information, making it impossible to consistently achieve higher-than-average returns through informed trading. It influences the assumptions of both CAPM and APT.
- Factor Sensitivities (Beta_{ij})** : In APT, factor sensitivities represent how an asset responds to each specific systematic risk factor. Each beta (beta_{ij}) measures the asset's sensitivity to a particular macroeconomic factor.
- Market Portfolio** : The market portfolio in CAPM is a theoretical portfolio that includes all assets in the market, weighted by their market values. CAPM assumes that investors hold this portfolio along with the risk-free asset.
- Market Risk Premium** : The market risk premium is the excess return expected from the overall market compared to the risk-free rate. It is a key input in both CAPM and APT formulas, representing compensation for taking on systematic risk.
- Rational Expectations** : Rational expectations assume that individuals form their expectations about future asset prices based on all available information. Both CAPM and APT rely on the assumption of rational expectations for the proper functioning of markets and pricing models.
- Risk-Free Rate** : The risk-free rate represents the theoretical return on an investment with zero risk. It is typically approximated using the yield on government bonds, such as Treasury bills.
- Systematic Risk (Beta, β)** : Beta is a measure of an asset's sensitivity to changes in the market portfolio. A beta of 1 indicates the asset moves in line with the market, while a beta less than 1 implies lower volatility, and a beta greater than 1 indicates higher volatility.
- Unsystematic Risk** : Also known as specific or idiosyncratic risk, unsystematic risk is the risk that can be diversified away in a well-constructed portfolio. APT assumes that investors can eliminate unsystematic risk through proper portfolio diversification.

20.8 SOME USEFUL BOOKS/ARTICLES

Ross, Stephen A. Randolph W. Westerfield, and Jeffrey Jaffe (2019). *Corporate Finance: Core Principles and Applications* McGraw Hill

Ross A.S, The Arbitrage Theory of Capital Asset Pricing, Journal of Economic Theory, Volume 13, Issue 3, 1976, Pages 341-360, ISSN 0022-0531.

Jensen, Michael C. and Black, Fischer and Scholes, Myron S. and Scholes, Myron S., The Capital Asset Pricing Model: Some Empirical Tests, in Michael C. Jensen (ed), *Studies In The Theory Of Capital Markets*, Praeger Publishers Inc., 1972.

20.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Single-Factor Model, homogeneous Expectations, Static Risk, Risk-Free Rate Assumption.
- 2) Perfect Markets, homogeneous Assets, normal Distribution of Returns, constant correlations, investor rationality.

Check Your Progress 2

- 1) Refer to Sub-section 20.3.1
- 2) $E(R_i) = R_f + \beta_{i1} \times \lambda_1 + \beta_{i2} \times \lambda_2 + \dots + \beta_{ik} \times \lambda_k + \epsilon_i$

Investors cannot consistently earn risk-free profits by exploiting price discrepancies between assets. Rationality is crucial for the proper functioning of arbitrage and for ensuring that asset prices align with the expectations implied by the model.

- 3) (CAPM) relies on a single systematic risk factor, which is market risk. (APT) allows for the consideration of multiple systematic risk factors.

Check Your Progress 3

- 1) Data Limitations, model misspecification, time-varying betas, factor redundancy, market efficiency concerns.
- 2) Refer to Sub-section 20.5.2

UNIT 21 PRICING OF DERIVATIVES

Structure

- 21.0 Objectives
- 21.1 Introduction
- 21.2 Derivatives: Basic Concepts
- 21.3 Types of Derivatives
 - 21.3.1 Forward Contract
 - 21.3.2 Futures
 - 21.3.3 Options
 - 21.3.4 Swaps
 - 21.3.5 Put – Call Parity
- 21.4 Models of Derivative Pricing
 - 21.4.1 Binomial Option Pricing Model
 - 21.4.2 The Black Scholes Formula
 - 21.4.3 Relation between the Binomial Options Pricing Model & Black-Scholes Model
- 21.5 Market of Derivatives in India
- 21.6 Let Us Sum Up
- 21.7 Key Words
- 21.8 Some Useful Books
- 21.9 Answers/Hints to Check Your Progress Exercises

21.0 OBJECTIVES

After going through this unit, you will be able to:

- explain the concept of a derivative;
- describe the various types of derivatives;
- discuss various models of derivative pricing; and
- explain different models of derivative pricing.

21.1 INTRODUCTION

In recent years, derivative products such as futures and options have gained significant prominence in the Indian stock markets, playing crucial roles in price discovery, portfolio diversification, and risk hedging. Notably, the National Stock Exchange (NSE) has witnessed a remarkable surge in trading volumes within its derivative markets, surpassing the turnover observed in the spot markets.

The notional turnover (i.e. Sum of (the strike price of each contract traded today multiplied by the number of the underlying asset in each contract) of futures and options on the National Stock Exchange of India Ltd., recognized as the world's largest derivatives bourse in terms of the number of contracts traded, has experienced a notable increase over the past year. As of 2nd January, 2024, based on the 10-day moving average, the turnover reached \$5.04 trillion, marking a significant rise from \$2.5 trillion recorded at the close of 2022, according to information gathered by Bloomberg.

Given this substantial growth, understanding the intricacies of derivative instruments becomes increasingly vital for market participants. These financial tools continue to play a pivotal role not only in enhancing price discovery mechanisms but also in providing avenues for portfolio diversification and effective risk management. The persistently higher turnover in the derivative markets, as compared to the cash segment, underscores the significance of these instruments in the contemporary Indian financial landscape. You have already studied about basic concepts of derivatives in Unit 13 where we discussed about characteristics of derivatives and types of derivatives. In this unit, we will extend our discussion and examine various models of derivative pricing.

21.2 DERIVATIVES: BASIC CONCEPTS

We have discussed about derivatives and related concepts in Unit 13. We already know that “Derivative” implies that its value is not independent but rather “derived” from an underlying asset, which can range from financial instruments like securities and commodities to various other entities. Derivatives encompass contracts such as forwards, futures, options, or hybrids, with predetermined durations, linked to the value of specified real or financial assets or an index of securities. Referred to as “deferred delivery of deferred payment instruments,” derivatives resemble securitized assets but lack the backing obligations from the original issuer of the underlying assets.

Derivative contracts have fixed expiry periods, typically ranging from 3 to 12 months, starting from the contract commencement date. The contract value depends on both the expiry period and the underlying asset's price. For instance, a farmer concerned about potential declines in wheat prices may enter into a derivative contract with a merchant, securing a set price (exercise price) for the crop at a future date. The derivative's value fluctuates based on the underlying asset's market value.

Commodity derivatives involve underlying assets like coffee, wheat, or precious stones, while financial derivatives involve assets such as debt instruments, currency, or equity shares. Derivatives can be exchange-traded or customized over-the-counter (OTC) based on standardization or negotiation with the involved parties.

To illustrate, a farmer anticipating a soybean production of 150 quintals in three months may either engage in an OTC derivatives contract with a food

merchant or purchase a standardized wheat contract on a commodities exchange, leaving some quantity uncovered for price fluctuations. Exchange-traded derivatives offer advantages like lower transaction costs and reduced default risks.

The economic functions of derivatives markets include risk transfer from risk-averse to risk-oriented individuals, price discovery for future and current prices, stimulation of entrepreneurial activity, increased market trading volumes due to the participation of risk-averse individuals, and promotion of savings and long-term investments.

Check Your Progress 1

- 1) Explain the concept of a derivative.

.....
.....
.....

- 2) List the various functions of derivative markets.

.....
.....
.....

21.3 TYPES OF DERIVATIVES

In the realm of financial markets, a diverse array of derivatives is actively traded. We have discussed about types of derivatives in Unit 13. In this unit, we will revisit them briefly. Since these financial instruments play a significant role in financial market, we will focus on following derivatives in this unit:

- 1) Forward Contracts
- 2) Futures Contracts
- 3) Options
- 4) Swaps

21.3.1 Forward Contract

In financial markets, the ancient practice of Forward Contracts (forwards) has endured across various nations, including India, standing out as the most straightforward derivative. You must have already understood from Unit 13 that a forward contract is a bespoke **agreement between two entities, culminating in settlement on a predetermined future date at an agreed-upon current price.** This bilateral contract involves one party assuming a long position, committing to buy the underlying asset on a specified future

date for a predetermined period. Simultaneously, the other party assumes a short position, agreeing to sell the asset on the same date at the same price.

For example: Let us consider a scenario where two parties enter into a contract to buy and sell 100 shares of Reliance at Rs 850 per share, two months from the contract date. The terms, including the product (Reliance shares), quantity (100 shares), price (Rs. 850 per share), and delivery time (2 months from the contract date), are predetermined.

However, forward contracts carry default risks. For instance, if the market price of Reliance shares significantly rises, the seller may prefer selling in the open market rather than honouring the contract, leading to a potential default. Similarly, if prices drop, the buyer may default, opting to purchase at the lower market rate. This mutual exposure to default risk is a drawback of customized forward contracts.

In India, forward markets, like those in the foreign exchange market, offer flexibility in negotiating terms such as price, quantity, quality, and delivery time and place. However, due to customization, they suffer from poor liquidity and credit risks.

To calculate payoffs from forward contracts, consider an example where two parties agree to exchange 500,000 barrels of crude oil at US \$ 42.08 per barrel in three months. A forward contract involves variables like the underlier, notional amount (n), delivery price (k), and settlement date. The payoffs are linear, determined by the market value formula: $n(s - k)$. Here s stands for the price prevailing in the spot market. In a forward contract, the “underlier” refers to the asset or security that the contract is based on. It is the specific item or financial instrument that the buyer and seller agree to exchange at a future date. The underlier could be a commodity, currency, stock, bond, or any other tradable asset.

For example, if two parties enter into a forward contract to buy and sell 100 shares of Company X’s stock at a predetermined price three months from now, the stock of Company X is the underlier in that forward contract. In essence, the underlier is the subject of the forward contract, and its value or performance will determine the financial outcomes and obligations of the contracting parties at the time of settlement.

Forward contracts may be cash-settled, meaning the underlier and payment don’t physically exchange hands. Instead, a single payment is made based on the market value at settlement. The bid and offer forward prices, influenced by factors like the time value of money and market expectations, are crucial. The graph of forward prices for various maturities is known as a forward curve.

21.3.2 Futures

In financial markets, Futures Contracts represent legally binding agreements to buy or sell an underlying security on a predetermined future date. These

contracts are organized and standardized concerning quantity, quality (for commodities), delivery time, and place, allowing for settlement on any future date. As you already know from Unit 13, the contract has a specified expiry date and settlement can occur either through the delivery of the underlying asset or in cash.

In contrast to Forward Contracts, Futures Contracts are typically traded on exchanges. The organized nature of these markets ensures high liquidity, mitigating the liquidity issues often present in the forward market. A clearing corporation or house in futures markets acts as the counterparty for all trades, providing an unconditional guarantee for settlement and maintaining financial integrity.

For instance, consider the scenario where parties A and B enter into a contract to buy and sell Reliance shares. If this contract is facilitated through an exchange, with a clearing corporation providing settlement guarantees, it is termed a futures contract.

The transition from a standardized and exchange-traded forward contract to a futures contract is evident, with futures offering advantages such as transparency and robust risk management. Settlement of futures contracts can occur through physical delivery of assets or in cash, with clear specifications outlined by the exchanges.

In a futures transaction, two parties negotiate through brokerage firms that hold a “seat” on the exchange for that particular contract. Legally structured as two contracts, each with the clearinghouse, this arrangement eliminates direct legal obligations between the parties. The margining process, involving daily profit or loss calculations, ensures ongoing settlement, effectively eliminating credit risk for futures.

Futures contracts can take various forms, including Index Futures and Stock Futures. Index Futures are based on an underlying index, cash-settled and deriving value from the index itself. On the other hand, Stock Futures involve specific stocks as underliers, with single stock futures being cash-settled.

21.3.3 Options

As we have discussed in Unit 13, Options is a contract, a subset of derivatives contracts that allows the buyer/holder the right (without the obligation) to buy/sell the underlying asset at a predetermined price within or at the end of a specified period. The buyer acquires this right from the seller, known as the writer, in exchange for a consideration called the premium. The seller is obligated to fulfil the terms of the contract when the buyer decides to exercise their right. Underlying assets for options can include securities, indices, or other financial instruments.

As per the Securities Contracts (Regulations) Act, 1956, options on securities are defined as contracts for the purchase or sale of the right to buy or sell securities in the future, encompassing various types like teji, mandi, galli,

put, call, or put and call in securities. We have already discussed about Call and Put options in Unit 13 in details.

A Call option provides the right to buy, while a Put option provides the right to sell. American options can be exercised anytime on or before the expiry date, while European options are exercisable only on the expiry date. The agreed-upon price at which the option can be exercised is called the Strike price or Exercise price. Unlike forward or futures contracts, acquiring an option involves a cost known as the premium.

For instance, consider a European exercise, a three-month put option on 100,000 barrels of Brent oil with a strike price of US \$45. This option gives the holder the right to sell 100,000 barrels of Brent oil at US \$45 per barrel three months from today.

An analogy to buying a television illustrates the concept further. If Mr. A pays Rs. 200-300 to reserve a television for two days with the option to buy later, he is akin to an option buyer. If he finds a better deal elsewhere, he may choose to let the option expire, akin to not exercising the right. The Rs. 200-300 paid serves as the price of the option.

Options contracts can be settled through the delivery of the underlying asset or in cash. Cash settlement in options involves paying/receiving the difference between the strike price and the price of the underlying asset at the time of expiry or exercise/assignment.

Index options contracts, based on an index, provide the right but not the obligation to buy/sell the underlying index on expiry. These are generally European-style options and are cash-settled on expiry.

In options markets, the exercise price determines whether a call option is in the money (current price exceeds exercise price) or out of the money (current price is less than exercise price). The premium is the price paid by the buyer to the seller, and a covered option is written against an owned asset, while a naked option is written without owning the asset.

21.3.4 Swaps

In Unit 13, we discussed that a **swap** is a contractual agreement between two parties to exchange assets, financial obligations, or cash flows over a specified period at predetermined intervals. Various forms of swaps have emerged in the market, including interest rate swaps, coupon swaps, basis rate swaps, bond swaps, substitution swaps, Intermarket spread swaps, swaps with timing mismatches, swaps with options like payoffs, and currency swaps. Among these, interest rate swaps and currency swaps are the most commonly employed. We have discussed different forms of Swaps in Unit 13. The most prevalent type of swap is the “plain vanilla” interest rate swap. In this arrangement, a company commits to paying cash flows equivalent to interest at a predetermined fixed rate on a notional principal for a specified period. In return, it receives interest at a floating rate on the same notional

principal for the same duration, often linked to the London Interbank Offer Rate (LIBOR).

An interest rate swap involves the exchange of one stream of interest obligations between two parties, with a specific maturity on a notional principal amount. The notional principal serves as a reference amount for interest calculations, with no actual exchange of principal occurring. Maturities for these transactions can range from less than a year to over 15 years, with the majority falling within the two to 10-year period.

On the other hand, a currency swap is a contractual agreement where foreign currencies are exchanged in the spot market, with a simultaneous commitment to reserve the transaction in the forward market. The exchange rate and timing of the forward market transaction are predetermined at the time of the swap. In essence, a currency swap involves swapping both principal and interest in one currency for equivalent amounts in another currency. Upon maturity, the principal amounts are swapped back.

For instance, a company facing exchange rate risk due to borrowing rupees at a fixed interest rate can mitigate this risk by entering a currency swap. The company sets up a contract to receive rupees at a fixed rate in exchange for dollars at either a fixed or floating interest rate. This strategy allows the company to manage both interest rate and exchange rate risks effectively.

The derivative market has achieved considerable success, attracting diverse traders due to its liquidity. **Traders in this market fall into three broad categories: hedgers, speculators, and arbitrageurs.**

- **Hedgers:** This group employs forwards, futures, and options to mitigate the risk associated with potential future movements in market variables. By utilizing derivative instruments, hedgers seek to safeguard themselves against adverse market fluctuations.
- **Speculators:** Traders in this category use derivatives to speculate on the future direction of market variables. Unlike hedgers who aim to reduce risk, speculators actively bet on market movements, seeking profit from price fluctuations.
- **Arbitrageurs:** Arbitrageurs engage in taking offsetting positions in two or more instruments simultaneously to lock in a profit. They capitalize on price differentials between related assets in different markets, exploiting temporary discrepancies for financial gain.

21.3.5 Put – Call Parity

In financial mathematics, put–call parity establishes a relationship between the prices of European call and put options with identical strike prices and expiry dates. This relationship relies on the absence of arbitrage in the market.

For example, considering stock options, let's denote the underlier value at expiration as S and the strike price as K . If S is less than or equal to K , the put option has a value of $(K-S)$, and the share has a value of S . If S is greater than or equal to K , the put option has a value of 0, and the share has a value of S .

Considering two portfolios — one with a put option and a share, and the other with a call option and K bonds — the values at expiration are compared. The portfolios have equal values at any time t before expiration. This equality prevents arbitrage opportunities, establishing the following relationship between the instruments' values at time t :

$$C(t) + K \times B(t, T) = P(t) + S(t)$$

Where:

- $C(t)$ is the time- t value of the call
- $P(t)$ is the time- t value of the put
- $S(t)$ is the time- t value of the share
- K is the strike price
- $B(t, T)$ is the time- t value of a bond that pays at T .

Put-call parity implies:

- 1) **Equivalence of calls and puts:** Calls and puts can be used interchangeably in any delta-neutral portfolio. If d is the call's delta, then buying a call and selling d shares of stock is equivalent to buying a put and buying $1-d$ shares of stock. This equivalence is crucial in options trading.

Parity of implied volatility: In the absence of dividends or other costs of carry, the implied volatility of calls and puts must be identical, ensuring consistency in market expectations.

In summary, understanding the diverse roles of hedgers, speculators, and arbitrageurs, along with the principles of put-call parity, contributes to a comprehensive grasp of derivative markets and their underlying mathematical relationships.

Check Your Progress 2

- 1) Explain what is meant by a forward contract.

.....

- 2) Explain the workings of a futures contract.

.....
.....
.....
3) What is a swap?
.....
.....
.....

21.4 MODELS OF DERIVATIVE PRICING

Derivative pricing is a complex process as it involves determining the fair value of a derivative contract. The pricing of derivatives depends on various factors, including the type of derivative, the characteristics of the underlying asset, market conditions, and the methodologies used. This section discusses two popular derivative pricing models: **Binomial Option Pricing Model** and **Black-Scholes Option Pricing Model**.

21.4.1 Binomial Option Pricing Model

The binomial options pricing model, initially introduced by Cox, Ross, and Rubinstein in 1979, offers a versatile numerical approach to valuing options. This model employs a “discrete-time” framework to track the evolving price of the underlying financial instrument over time, facilitating option valuation through the application of the risk neutrality assumption.

Unlike other models, the binomial options pricing model excels in handling diverse conditions, including American options that can be exercised at any point and Bermudan options exercisable at various points. Its simplicity and mathematical feasibility make it easily implementable in software or spreadsheet environments. While it may be slower than the Black-Scholes model, practitioners often prefer it for its accuracy, especially in valuing longer-dated options and those associated with dividend payments.

For options involving multiple uncertainties (e.g., real options) or complex features (e.g., Asian options), lattice methods encounter challenges, making Monte Carlo option models more suitable. However, Monte Carlo simulation, although comprehensive, is computationally intensive and not the preferred choice when lattice methods or formulas are adequate.

The binomial pricing model operates within a “discrete-time framework,” utilizing a binomial lattice (tree) to trace the evolution of the option’s underlying variable across multiple time steps. Each node in the lattice represents a potential price of the underlying at a specific point in time, forming the basis for option valuation. The valuation process involves three key steps:

- 1) Price tree generation: Creating a tree to depict the evolution of the underlying asset's price.
- 2) Calculation of option value at final nodes: Determining the option's intrinsic (exercise) value at the expiration of the option.
- 3) Progressive calculation of option value at earlier nodes: Iteratively calculating the option value at each preceding node using the risk neutrality assumption, where the fair price today is equal to the discounted expected value of its future payoff. The "Binomial Value" at each node is determined by weighting the option values from the later two nodes by their respective probabilities of an up or down move and discounting them at the risk-free rate.

For an American option, which can be held or exercised before expiry, the value at each node is the maximum of the Binomial Value and the Exercise Value. The Binomial Value is computed using the formula:

$$\text{Binomial Value} = [p \times \text{Option up} + (1-p) \times \text{Option down}] \times \exp(-r \times t)$$

Here, p represents the probability of an up move in the underlying,

$(1-p)$ is the probability of a down move,

r is the risk-free rate,

and t is the time to expiration.

This process is repeated for each node, concluding at the valuation date, where the final result represents the option's value.

21.4.2 The Black-Scholes Option Pricing Model

The Black-Scholes model serves as a mathematical framework for evaluating the evolving prices of financial instruments, particularly stocks. Fischer Black and Myron Scholes developed the Black-Scholes formula in 1973, building upon earlier work by Edward Thorpe, Paul Samuelson, and Robert C. Merton. This model is widely used in practice to value call and put options. The model is based on the following assumptions:

- 1) The underlying share/stock does not generate dividends during the life of the call option.
- 2) Transaction costs are zero.
- 3) The risk-free rate of return is known and it remains constant during the life of the option.
- 4) The option can be exercised at the expiry only not before that. It means the option is a European not American.
- 5) Short selling is allowed and the short seller will get immediately the full cash proceeds of today's price for a security sold short.
- 6) Stock price moves randomly.

If the above assumptions are met, then as per the Black-Scholes Option Pricing Model, the value of a call option can be calculated using the following formula:

$$VCO = P[N(d_1)] - Ye^{-rt}[N(d_2)]$$

Such that

$$d_1 = \frac{\ln(P/Y) + [r + (\sigma^2/2)]t}{\sigma\sqrt{t}}$$

$$d_2 = d_1 - \sigma\sqrt{t}$$

where,

VCO = Value of a call option

P = Current price of the underlying share or stock

Y = Exercise/strike price of the option

r = Risk – free rate of return (should be continuously compounded)

t = Time to the expiry of the call option (should be an annual figure and used in fraction)

σ^2 = Annualised variance of the rate of return on the underlying stock or share

$N(d_1)$ and $N(d_2)$ represent the cumulative areas under a standard normal distribution i.e., a normal distribution with zero mean and standard deviation 1.

$\ln(P/Y)$ = Natural log of (P/Y)

$e = 2.7183$ (Approximately)

If we want to calculate the value of a put option, then the formula is

$$VPO = Ye^{-rt}[1 - N(d_2)] - P[1 - N(d_1)]$$

Example 21.1: Suppose we want to value a call option where the underlying asset is the share of Tata Consultancy Service. The exercise price of the option is ₹ 20 and the current price of the share is ₹ 25. Further, suppose that the treasury bills rate (a proxy used for risk-free rate) is 5%, the annualised variance of the returns on the share is 0.16, and the call option will expire at the end of 3 months. The value of the call option can be calculated as follows:

Solution

Step 1: Summarise the information

$P = ₹ 25$

$$Y = ₹ 20$$

$$r = 5\% \text{ or } 0.05$$

$$t = 3 \text{ months or } 3/12 = 0.25 \text{ years}$$

$$\sigma^2 = 0.16 \Rightarrow \sigma = 0.4$$

Step 2: Find out d_1 and d_2

$$d_1 = \frac{\ln(P/Y) + [r + (\sigma^2/2)]t}{\sigma\sqrt{t}}$$

$$d_1 = \frac{\ln\left(\frac{25}{20}\right) + \left[0.05 + \left(\frac{0.16}{2}\right)\right] 0.25}{0.4\sqrt{0.25}}$$

$$= \frac{\ln(1.25) + (0.13) \times 0.25}{0.4 \times 0.5}$$

Using Excel/Scientific calculator/log table, we can calculate the natural log of 1.25 i.e., $\ln(1.25) = 0.2231$

$$d_1 = \frac{0.2231 + 0.0325}{0.2} = \frac{0.2556}{0.2}$$

$$d_1 = 1.278$$

Therefore,

$$d_2 = d_1 - \sigma\sqrt{t} = 1.278 - 0.4\sqrt{0.25} = 1.278 - 0.2 = 1.078$$

Step 3: Find $N(d_1)$ and $N(d_2)$

Referring to the Standard Normal Distribution Table, we find the cumulative probabilities i.e.,

$$N(d_1) = N(1.278) = 0.8994$$

$$N(d_2) = N(1.078) = 0.8595$$

These cumulative probabilities can also be calculated using MS Excel. You should use the following Excel formula “= NORM.DIST(d_1 ,0,1, True)”.

Now, the value of the call option as per the Black-Scholes Option Pricing Model is

$$VCO = P[N(d_1)] - Ye^{-rt}[N(d_2)]$$

$$VCO = 25(0.8994) - 20 \times e^{-0.05 \times 0.25}(0.8595)$$

$$= 22.485 - 20 \times e^{-0.0125}(0.8595)$$

We can calculate the value of $e^{-0.0125}$ using the log technique or scientific calculators or in MS-Excel using the formula “= POWER (2.7183, -0.0125)”. We find that $e^{-0.0125} = 0.9875$. Thus,

$$VC0 = 22.485 - 20 \times 0.9875 \times (0.8595)$$

$$= 22.485 - 16.97513$$

$$VCO = 5.509875 \approx 5.5099$$

Thus, the call option value is ₹ 5.5099.

Suppose it was a **put option**, then

$$VPO = Ye^{-rt}[1 - N(d_2)] - P[1 - N(d_1)]$$

$$VPO = 20 \times 0.9875 \times [1 - 0.8595] - 25[1 - 0.8994]$$

$$= 2.7748 - 2.515$$

$$= 0.2598$$

Advantages & Disadvantages of the Black-Scholes Model:

Advantage: One major advantage of the Black-Scholes Model is its speed, allowing for the rapid calculation of a large number of option prices.

Limitation: The model's significant limitation is its inability to accurately price American-style options, as it only calculates prices at expiration and does not account for potential early exercise steps.

Assumptions underlying the Black-Scholes model include the absence of arbitrage opportunities, the possibility of short selling at all times, no transaction costs or taxes, perfect divisibility of securities, continuous trading, and no dividends paid by the underlying stock during the option's lifetime. Additionally, it assumes that the risk-free rate (r) and stock volatility (σ) are known over the option's lifetime.

21.4.3 Relation between the Binomial Options Pricing Model & Black-Scholes Model

Both the binomial and Black-Scholes models share a foundation rooted in the assumption that stock prices follow a stochastic process described by geometric Brownian motion. For European options, the binomial model approaches the Black-Scholes formula as the number of binomial calculation steps increases. In essence, the Black-Scholes model for European options can be seen as a special case of the binomial model with an infinite number of steps, representing a continuous process approximation. Therefore, the binomial model offers discrete approximations to the continuous process underlying the Black-Scholes model.

- 1) How will you determine the value of call and put options using the Black- Scholes Option Pricing Model?

.....

- 2) The current price is ₹ 35, the exercise price is ₹ 35, the risk-free rate of interest is 10%, the volatility of returns of the underlying asset is 20% and the time to expiry is 1 year. Calculate the value of the call option using the Black-Scholes Option Model. What would be your answer if it were a put option, not a call option? (**Hint:** Refer to example 4).

.....

Note: Students can use an online calculator (scan the QR code) to value options as per the Black-Scholes Option Pricing Model.



21.5 MARKET OF DERIVATIVES IN INDIA

India has two types of derivative markets: the exchange-traded market and the over-the-counter (OTC) market. In the exchange-traded market, standardized contracts are traded on an exchange. In contrast, the OTC market is decentralized, with contracts negotiated directly between two parties. SEBI is the regulator of the derivative market.

Trade in equity and index derivatives

If you want to trade in equity and index derivatives, you can do so through the National Stock Exchange (NSE) or Bombay Stock Exchange (BSE). The derivative market in India started in the early 1990s. The first derivatives contract was launched on the Bombay Stock Exchange (BSE) in June 2000. Since then, the derivative market in India has grown exponentially¹. There

¹ <https://www.bsebti.com/blog/derivative-market-in-india-meaning-types-participants-differences/#:~:text=Difference%20Between%20Cash%20and%20Derivative%20Market&text=The%20derivative%20market%20in%20India,in%20India%20has%20grown%20exponentially.>

are two types of markets – the cash market and the derivative market. The difference between cash and derivative market is that transactions are settled on the spot in the cash market. In contrast, in the derivative market, transactions are settled at a future date. The participants in the cash market are known as spot traders, while the participants in derivative market are known as futures traders. Spot trading is done in the physical form of commodities, whereas futures trading is done via derivatives contracts.

The National Stock Exchange of India Limited (NSE) commenced trading in derivatives with the launch of index futures on June 12, 2000². The futures contracts are based on the popular benchmark Nifty 50 Index. The Exchange introduced trading in Index Options (also based on Nifty 50) on June 4, 2001. NSE also became the first exchange to launch trading in options on individual securities from July 2, 2001. Futures on individual securities were introduced on November 9, 2001. Futures and Options on individual securities are available on securities that meet the eligibility criteria stipulated by SEBI. The Exchange has also introduced trading in Futures and Options contracts based on Indices. Currently, Derivatives on NIFTY 50, NIFTY Bank, NIFTY Financial Service, NIFTY Midcap Select and NIFTY Next 50 are available for trading. Since the launch of the Index Derivatives on the popular benchmark Nifty 50 Index in 2000, the National Stock Exchange of India Limited (NSE) today have moved ahead with a varied product offering in equity derivatives. The Exchange currently provides trading in Futures and Options contracts on – 5 major indices and more than 180 securities.

In India, both the BSE and NSE provide derivatives trading facilities. The NSE's derivatives platform is known as Nifty Futures and Options (F&O). The BSE offers F&O trading on its own exchange and also facilitates trading on the NSE's exchange.

Trading in Commodity Derivatives

If you want to trade in commodities like crude oil, gold, metals, etc., you can do so through commodity exchanges like the Multi Commodity Exchange (MCX) or the National Commodity and Derivatives Exchange (NCDEX) both under the ownership of Ministry of Finance, Government of India. Conversely, if you want to trade in currencies, Metropolitan Stock Exchange of India (MSE) Limited under the ownership of Ministry of Finance, Government of India, simplifies that.

21.6 LET US SUM UP

This unit builds upon the topics explored in units 13, 18 and 19, focusing on assets and their pricing. The discussion centres around a specific type of asset whose value is intricately tied to and influenced by the price movements of

² <https://www.nseindia.com/products-services/about-equity-derivatives>

an underlying asset. The unit commences with an introduction to the fundamental concept and rationale behind derivatives. Subsequently, it delves into an extensive examination of various types of derivatives, encompassing forward contracts, futures, and swaps.

The unit then shifts its focus to the pricing mechanisms of derivatives, providing insights into both the binomial model and the Black-Scholes formula. A comparative analysis is conducted to highlight the distinctions between these two models.

21.7 KEY WORDS

Arbitrageurs	: Investors engaging in arbitrage to exploit price discrepancies in securities to earn riskless returns.
Call Option	: A financial contract providing the holder the right to purchase an asset at a predetermined price within a specified timeframe.
Forward Contract	: A mutual agreement between two parties to exchange an asset for cash at a predetermined future date, with the current price fixed at the time of agreement.
Hedge	: An investment made to offset potential losses in another investment, creating a balance between the two.
Interest Rate Future	: Futures contracts based on fixed-income securities like treasury bills, bonds, and CDs.
Interest Rate Options	: Options contracts related to fixed-income securities such as treasury bills.
Interest Rate Parity	: The concept suggesting that the difference in interest rates between two countries equals the disparity between the forward and spot rates of those respective countries.
Long Position	: The position of an investor entitled to receive an asset in the future, representing the buyer's role in the security.
Margin	: A portion of the transaction value that the customer must pay to initiate a transaction, with the remaining amount often borrowed from the broker. Two types of margins exist: initial margin and maintenance margin.

Option Contract	: A contractual agreement providing the right to buy or sell an asset at a predetermined price on a future date, with the option buyer having the discretion to exercise this right.
Put Option	: An option granting the holder the right to sell an asset at a fixed price within a specified period.
Short Position	: The position of an investor obligated to deliver an asset in the future, indicating the seller's role in the security.
Spot Rate	: The exchange rate applicable to the immediate delivery of currency, typically settling two days after the trade date.
Swap Contract	: An agreement between two parties (counterparties) to exchange assets or a series of cash flows over a specified period at predetermined intervals.

21.8 SOME USEFUL BOOKS

Bhole, L M . (2004) *Financial Institutions and Markets 4th edition*, TataMcGraw- Hill, New Delhi:

Dothan, U. (1990), *Prices in Financial Markets*, Oxford University Press, New York

Hull, John, (2002) *Options, Futures and other Derivatives* Prentice-Hall of India, New Delhi.

“Options, Futures, and Other Derivatives” by John C. Hull

21.9 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Derivatives encompass contracts such as forwards, futures, options, or hybrids, with predetermined durations, linked to the value of specified real or financial assets or an index of securities.
- 2) Risk transfer from risk-averse to risk-oriented individuals, price discovery for future and current prices, stimulation of entrepreneurial activity, increased market trading volumes due to the participation of risk-averse individuals, and promotion of savings and long-term investments.

Check Your Progress 2

- 1) Agreement between two entities, culminating in settlement on a predetermined future date at an agreed-upon current price.
- 2) Futures Contracts are typically traded on exchanges. In a futures transaction, two parties negotiate through brokerage firms that hold a “seat” on the exchange for that particular contract.
- 3) A swap is a contractual agreement between two parties to exchange assets, financial obligations, or cash flows over a specified period at predetermined intervals.

Check Your Progress 3

- 1) Refer to Sub-section 21.4.2
- 2) The current price is ₹ 35, the exercise price is ₹ 35, the risk-free rate of interest is 10%, the volatility of returns of the underlying asset is 20% and the time to expiry is 1 year. Calculate the value of the call option using the Black-Scholes Option Model. What will be your answer if it were a put option not call option?

Step 1: Summarise the information

$$P = ₹ 35$$

$$Y = ₹ 35$$

$$r = 10\% \text{ or } 0.10$$

$$t = 1 \text{ year}$$

$$\sigma^2 = 0.04 \Rightarrow \sigma = 0.20$$

Step 2: Find out d_1 and d_2

$$d_1 = \frac{\ln(P/Y) + [r + (\sigma^2/2)]t}{\sigma\sqrt{t}}$$

$$d_1 = \frac{\ln\left(\frac{35}{35}\right) + \left[0.10 + \left(\frac{0.04}{2}\right)\right] 1}{0.20\sqrt{1}}$$

$$= \frac{\ln(1) + (0.12)}{0.20}$$

Using Excel/Scientific calculator/log table, we can calculate natural log of 1
i.e., $\ln(1) = 0$

$$d_1 = \frac{0 + 0.12}{0.20} = \frac{0.12}{0.20}$$

$$d_1 = 0.6$$

Therefore,

$$d_2 = d_1 - \sigma\sqrt{t} = 0.6 - 0.20\sqrt{1} = 0.6 - 0.20 = 0.4$$

Step 3: Find $N(d_1)$ and $N(d_2)$

Referring the Standard Normal Distribution Table, we find the cumulative probabilities i.e.,

$$N(d_1) = N(0.6) = 0.7257$$

$$N(d_2) = N(0.4) = 0.6554$$

These cumulative probabilities can also be calculated using MS-Excel. You should use the following excel formula “= NORM.DIST ($d_1, 0, 1$, True)”.

Now, the value of the call option as per the Black-Scholes Option Pricing Model is

$$VCO = P[N(d_1)] - Ye^{-rt}[N(d_2)]$$

$$VCO = 35(0.7257) - 35 \times e^{-0.10}(0.6554)$$

$$= 35 \times 0.7257 - 35 \times 0.9048 \times 0.6554 = 4.644$$

Note: We have calculated the value of $e^{-0.10}$ using log technique or scientific calculators or in MS-Excel using the formula “= POWER (2.7183, -0.10)”. We find that $e^{-0.10} = 0.9048$.

Thus, the call option value is ₹ 4.644.

Suppose that it were a **put option**, then

$$VPO = Ye^{-rt}[1 - N(d_2)] - P[1 - N(d_1)]$$

$$VPO = 35 \times 0.9048 \times [1 - 0.6554] - 35[1 - 0.7257]$$

$$= 11.91024 - 9.6005$$

$$= 1.312$$

UNIT 22 CORPORATE FINANCE

Structure

- 22.0 Objectives
- 22.1 Introduction
- 22.2 Sources of Finance
- 22.3 Capital Structure
 - 22.3.1 Components of Capital Structure
 - 22.3.2 Factors Influencing Capital Structure
- 22.4 Working Capital Management
 - 22.4.1 Importance of Working Capital
 - 22.4.2 Factors Determining Working Capital
 - 22.4.3 Methods of Making Working Capital Forecast
 - 22.4.4 Sources of Working Capital
 - 22.4.5 Approaches for Determining the Financial Mix
- 22.5 Dividend Policy
- 22.6 Capital Budgeting
 - 22.6.1 Importance of Capital Budgeting
 - 22.6.2 Kinds of Capital Investment Proposals
 - 22.6.3 Factors Affecting Capital Investment Decision
 - 22.6.4 Computation of Capital Investment
 - 22.6.5 Forecasting Future Costs
- 22.7 Let Us Sum Up
- 22.8 Key Words
- 22.9 Suggested Readings
- 22.10 Answers/Hints to Check Your Progress Exercises

22.0 OBJECTIVES

After going through this unit, you will be able to:

- explain the concept of corporate finance;
- examine the concept of capital structure and its theories;
- explain the techniques of capital budgeting;
- list and describe the sources of working capital and its management; and
- describe the theories of dividend policy and the factors affecting dividend decision.

22.1 INTRODUCTION

When we think of Corporate Finance, the first thing that comes to our mind is **managing money** in a business, from raising funds to investing them wisely. Corporate finance is the area of finance that deals with the sources of funding, and the capital structure of corporations, the actions that managers take to increase the value of the firm to the shareholders, and the tools and analysis used to allocate financial resources. The primary goal of corporate finance is to maximise or increase shareholder value.

Correspondingly, corporate finance comprises two main sub-disciplines. Capital budgeting is concerned with the setting of criteria about which value-adding projects should receive investment funding, and whether to finance that investment with equity or debt capital. Working capital management is the management of the company's monetary funds that deal with the short-term operating balance of current assets and current liabilities; the focus here is on managing cash, inventories, and short-term borrowing and lending (such as the terms of credit extended to customers).

In this unit, we will study the diverse sources of finance, the Capital structure of a company, different techniques of capital budgeting, working capital management, and dividend policy.

22.2 SOURCES OF FINANCE

Finance is especially important for every business, serving as a fundamental catalyst for its optimal functioning. The adequacy of financial resources is paramount, as it directly impacts the efficiency and effectiveness of business operations. In this context, the financial needs of a business can be categorised into 2 distinct classes:

- 1) Short-term financial requirements.
- 2) Long-term financial requirements.

Short-term financial requirements of a business necessitate essential short-term funds, typically covering a period of up to twelve months. These funds are procured from avenues capable of promptly extending financial aid for a brief period and at a reasonable cost. Frequently, securing these funds is facilitated through short-term loans or the process of bill discounting offered by commercial banks.

Long-term financial needs of a business that last for more than a year are met by long-term funds. These funds can be divided into several types:

- i) Intermediate or medium-term funds
- ii) Long-term funds.

The mentioned group refers to money needed for more than 5 years. Businesses get this money from places that can give them financial help for a long time. These places might include selling shares, debentures, and getting loans from special money-related organisations. It is worth noting that regular banks have started to offer medium-term and long-term funding options to businesses. They might do this alone or with other special financial organisations like the Industrial Finance Corporation of India and the Industrial Credit and Investment Corporation of India. This is done to meet the different money needs of businesses.

Classification of Sources of Finance

The sources of finance can be classified into two categories: internal sources and external sources. External sources can be (i) **loans** or (ii) **equities and bonds or debentures**. Equities are shares, and give a share of ownership to the holder while bonds or debentures promise a steady stream of payments to the holder without conferring ownership rights. Loans are borrowed external funds. Internal sources are sources of funds that are from within the firm. The two principal internal sources of funds are depreciation and retained earnings. We elaborate on this below:

External Sources of Funds

1) Issue of Shares

Share represents ownership and voting rights in a company. The holder of a share is termed a shareholder, receiving dividends from the company as a return for their investments. However, dividend payments are not obligatory, and the company's Board of Directors has the authority to propose dividends. While the Board's recommendation can be altered by shareholders, it cannot be increased beyond the initial proposal.

Kinds of Shares

Publicly traded companies in India can issue two types of shares:

a) Preference Shares

Preference shares offer specific preferential rights over other share classes, including i) Guaranteed fixed dividends, which can be a fixed amount or rate. ii) Priority in capital repayment during company liquidation, preceding other share classes. Preference shares might be cumulative or non-cumulative, participating, or non-participating, and redeemable or irredeemable. Cumulative preference shares accumulate unpaid dividends until cleared. Non-cumulative shares lose dividend rights if there is no profit. Participating shares share surplus profits after fixed dividend payments to equity shareholders. Non-participating shares lack this privilege. Redeemable preference shares can be reclaimed during the company's lifespan, while irredeemable ones can only be redeemed during liquidation.

b) Equity Shares

These shares lack preferential rights, ranking below preference shares for dividend and capital repayment during liquidation. Dividend rates are variable, depending on profits and directors' intent. Equity shareholders have voting power at company general meetings and are drawn to the potential for returns despite the risk.

2) Issue of Debenture

A debenture may be a record issued by a company as proof of an obligation due from the company with or without a charge on the assets of the company. It could be a certificate issued by a company beneath its seal recognising an obligation due by it to its holders.

Kinds of Debentures

Debentures represent a form of debt for a company, favoured for their avoidance of equity dilution, serving as a loan mechanism for a company's long-term profitability and expansion. Familiarity with the several types of debentures in the market is crucial when considering which to invest in.

- a) **Convertible Debentures:** Among the diverse debenture types, convertible debentures stand out with their unique feature – the ability to transform into shares or stocks at a designated point in time or upon notification by the company. Despite offering lower interest rates compared to stocks, these debentures prove highly advantageous.
- b) **Partially Convertible Debentures:** Debentures that can be converted into shares up to a specific limit or percentage are categorised as partially convertible debentures. This hybrid form leaves a portion as debentures even after partial conversion, while the rest becomes part of the company's shares.
- c) **Non-Convertible Debentures:** Representing the fundamental type, non-convertible debentures lack the option for conversion into stocks throughout their existence after issuance.
- d) **Registered Debentures:** This type involves the transfer of debentures with well-documented and secure records, ensuring a tamper-proof transaction. Registered debentures are considered safer due to the reduced risk of fraud, in contrast to the bearer debentures discussed below.
- e) **Bearer Debentures:** Unregistered and transferable without mandatory documentation, bearer debentures can be delivered after purchase without the need for proof or records. This type is more susceptible to tax evasion and fraud, lacking a third-party involvement in the transaction.

- f) **Secured Debentures:** Resembling an alternative to loans, secured debentures require collateral for fund generation. In the event of the firm settling its debts upon closure, secured debenture holders receive priority payment.
- g) **Unsecured Debentures:** Requiring no collateral, unsecured debentures are less favoured during payments compared to secured debentures.
- h) **Redeemable Debentures:** Purchased for a predetermined period, redeemable debentures are repaid by the end of this specified period.
- i) **Irredeemable Debentures:** Also known as perpetual debentures, this type lacks a fixed redemption period for the invested amount.

Internal Sources of Funds

As its name suggests, these are funds generated internally by the company. The two most critical sources of inside financing are depreciation and retained earnings. These are:

- a) **Depreciation:** Depreciation is a source of internal finance through its non-cash item in books, but it is a recognition of the value of the asset which will reduce each year and attempting to identify the amount of cash to set aside when the asset is no longer in use to us. It is also being considered as an expense to reduce our taxable income.
- b) **Retained Earnings:** It is that part of the profit which is not distributed among shareholders as a dividend. These earnings are kept by the company itself for the future requirement of funds. It is additionally known as 'Inside Financing' or 'Ploughing back of Profits.'
- c) **Loan Financing:** Loan financing refers to the acquisition of funds by individuals, businesses, or governments through borrowing from a lender. In this financial arrangement, a borrower receives a sum of money from the lender with the commitment to repay the amount, usually with interest, over a specified period. Loan financing can serve various purposes, including personal expenses, business investments, or major capital projects. Lenders may include banks, financial institutions, or private entities, and the terms of the loan agreement typically outline the interest rate, repayment schedule, and any collateral requirements.

Check Your Progress 1

- 1) List several types of shares.

.....

.....

.....

- 2) Explain depreciation as a source of finance.

.....
.....
.....

- 3) Differentiate between secured & unsecured debentures.

.....
.....
.....

22.3 CAPITAL STRUCTURE

As discussed in the previous section, there are different sources of finance which can be used to finance a business. Capital structure is nothing but the composition of capital. Capital Structure refers to the composition and proportion of various sources of funds that a company employs to finance its operations, investments, and other financial activities. It represents the way a company combines equity (ownership capital) and debt (borrowed capital) to support its business activities. The capital structure decision involves determining the mix of equity and debt that best aligns with the company's financial goals, risk tolerance, and overall strategy. In this section, we will study the factors that influence the capital structure and what are the theories that have been developed to understand the capital structure.

22.3.1 Components of Capital Structure

Components of a capital fund of a company may be classified based on their sources. Hence, they can be majorly divided into two categories:

- a) **Owner's capital:** Owner's capital, also known as owner's equity or shareholder's equity, refers to the portion of a company's assets that belongs to its owners or shareholders. It represents the residual interest in the company's assets after deducting liabilities. In other words, the owner's capital is the net value of a company's assets that remains after satisfying all its debts and obligations. The owner's capital consists of Equity Shares, Preference Shares and Retained Earnings.
- b) **Borrowed Capital:** Borrowed capital, also known as debt capital, refers to funds that a company raises by borrowing money from external sources, such as financial institutions, investors, or bondholders. Borrowed capital is a debt obligation as it creates a legal obligation for the company to repay the borrowed funds along with interest within a specified period. The borrowed capital consists of debentures, term loans, lease financing and short-term borrowings.

22.3.2 Factors Influencing Capital Structure

- a) **Business Risk:** The nature and level of a company's business risk can influence its capital structure. Companies operating in highly volatile or cyclical industries may opt for lower debt levels to avoid the added financial risk associated with interest payments during economic downturns.
- b) **Tax Considerations:** Interest payments on debt are often tax-deductible, making debt financing a more tax-efficient option than equity financing. Companies in higher tax brackets may be inclined to use debt to take advantage of tax benefits.
- c) **Cost of Capital:** Companies strive to minimise their overall cost of capital, which is the weighted average cost of debt and equity. Finding the optimal mix that minimises the cost of capital is a key consideration in determining the capital structure.
- d) **Market Conditions:** Economic and market conditions play a role in capital structure decisions. For instance, during periods of economic stability, companies may be more willing to take on debt at favourable interest rates. Conversely, during economic downturns, access to debt financing may be more challenging, influencing a shift toward equity financing.
- e) **Company Size and Life Cycle:** Smaller or newer companies may find it more challenging to access debt markets and may rely more on equity financing. As a company grows and establishes a record of accomplishment, it may have more options for debt financing.
- f) **Flexibility and Control:** Equity financing gives shareholders ownership of the company, but it also dilutes existing ownership. Some companies may prefer debt financing to maintain control and avoid dilution of ownership. Debt also provides more flexibility in adjusting capital structure in response to changing conditions.
- g) **Asset Structure:** The type of assets a company owns can impact its capital structure. Capital-intensive industries (e.g., manufacturing) may use more debt to finance their assets, while service-oriented industries may have lower debt levels.
- h) **Lender and Investor Preferences:** The preferences and requirements of lenders and investors in the capital markets can influence a company's choice of capital structure. Several types of lenders and investors may have varying expectations regarding risk, return, and the mix of debt and equity.
- i) **Regulatory Environment:** Regulatory constraints, including debt-to-equity ratios and other financial regulations, can influence a company's capital structure decisions.

- j) **Market Perception:** The market's perception of a company's financial health and risk profile can impact its ability to raise capital and the cost of that capital.

Check Your Progress 2

- 1) Differentiate between owner's capital and borrowed capital.

.....
.....
.....

- 2) Explain the factors affecting capital structure.

.....
.....
.....

22.4 WORKING CAPITAL MANAGEMENT

The concept of working capital is fundamental in financial management and represents the company's ability to meet its short-term financial obligations. It is calculated by subtracting current liabilities from current assets. Working capital is essential for the day-to-day operations of a business and maintaining a smooth cash flow.

There are two primary forms of working capital:

- I) Gross Working Capital:** Gross working capital refers to the total current assets of a company. These are the assets that are either in cash or expected to be converted into cash within the operating cycle, typically one year. Gross working capital includes the following components:

- 1) **Cash and Cash Equivalents:** This includes currency, bank balances, and other highly liquid assets.
- 2) **Accounts Receivable:** The money owed to the company by its customers for goods or services sold on credit.
- 3) **Inventory:** The stock of goods a company holds for sale, production, or consumption.
- 4) **Short-term Investments:** Investments made by the company in highly liquid financial instruments with maturities of up to one year.

- II) Net Working Capital:** Net working capital is derived by deducting current liabilities from current assets. It represents the excess of current assets over current liabilities and indicates the company's ability to cover its short-term debts.

22.4.1 Importance of Working Capital

Gross Working Capital:

- a) Gross working capital is significant as it provides an overview of a company's short-term resource base.
- b) It helps management understand the scale of current assets in comparison to current liabilities.
- c) A high gross working capital might indicate that the company is carrying excessive levels of current assets, which can lead to increased carrying costs and inefficiencies.
- d) Conversely, a low gross working capital might imply inadequate liquidity to fund short-term obligations.

Net Working Capital:

- a) Net working capital is a more focused metric as it reflects the excess or deficit of current assets after settling current liabilities.
- b) A positive net working capital is preferred as it demonstrates that a company has enough liquid resources to meet its short-term obligations.
- c) It can also signify good financial health and efficient working capital management.
- d) However, negative net working capital is not necessarily always a bad sign. Some industries, like retail, may have negative net working capital due to the quick turnover of inventory and immediate collection of cash from customers. It is crucial to assess negative net working capital in the context of the company's business model and industry norms.

22.4.2 Factors Determining Working Capital

Working capital, the difference between a company's current assets and current liabilities, is a crucial aspect of financial management. Several factors influence the working capital requirements of a business, and understanding these factors is essential for effective working capital management. Here are the key factors that determine working capital:

- 1) **Nature of Business:** The type of industry and the nature of the business significantly impact the working capital needs. For instance, manufacturing companies often require higher levels of working capital due to the need to maintain raw materials, work-in-progress, and finished goods inventory. On the other hand, service-based businesses may have lower working capital requirements since they typically have minimal inventory and faster cash conversion cycles.

- 2) **Business Cycle and Seasonality:** Companies experience fluctuations in sales and production throughout their business cycles. During peak seasons or periods of high demand, businesses may require additional working capital to finance increased inventory and accounts receivable. Conversely, during low-demand periods, working capital needs may decrease.
- 3) **Sales Volume and Growth Rate:** Companies with higher sales volume or experiencing rapid growth may need more working capital to support increased operations, expand inventory levels, and extend credit terms to customers.
- 4) **Credit Policy:** The credit policy adopted by a company, such as the length of credit given to customers, significantly impacts accounts receivable. A lenient credit policy may lead to higher accounts receivable and increased working capital needs, while a strict credit policy may affect sales but reduce working capital requirements.
- 5) **Supplier Payment Terms:** The terms and conditions negotiated with suppliers influence accounts payable. Longer payment terms can help conserve cash and reduce working capital requirements, but it may also affect supplier relationships.
- 6) **Inventory Management:** The efficiency of inventory management affects working capital. Companies with streamlined inventory control systems and faster inventory turnover require lower levels of working capital.
- 7) **Seasonal Inventory Needs:** Businesses with seasonal inventory requirements may need to maintain higher working capital during peak seasons to meet customer demands.
- 8) **Operating Efficiency:** Efficient operational practices can help optimise working capital. Streamlined processes, reduced lead times, and effective use of technology can enhance cash flow and reduce working capital needs.
- 9) **Economic Conditions:** The overall economic environment, including inflation, interest rates, and credit availability, can impact working capital requirements. Economic downturns may increase the need for working capital to weather challenging times.

22.4.3 Methods of Making Working Capital Forecast

Making accurate working capital forecasts is essential for effective financial planning and management. Various methods can be used to make these forecasts, depending on the availability of data, the complexity of the business, and the desired level of precision. Some common methods of making working capital forecasts include:

1) Estimating the Amount of Current Assets and Current Liabilities:

This method involves projecting the future levels of individual current assets and current liabilities based on historical data and anticipated business activities. The working capital forecast is then determined by subtracting the estimated total of current liabilities from the estimated total of current assets.

Example 22.1: ABC Manufacturing Company needs to forecast its working capital requirements for the next quarter. Based on historical data and sales projections, they estimate the following current assets and current liabilities:

Current Assets: Cash and Cash Equivalents: Rs.50,000, Accounts Receivable: Rs.120,000, Inventory: Rs.80,000, Prepaid Expenses: Rs.10,000, Total Current Assets: Rs.260,000

Current Liabilities: Accounts Payable: Rs.90,000, Short-Term Debt: Rs.30,000, Accrued Expenses: Rs.15,000, Taxes Payable: Rs.5,000, Total Current Liabilities: Rs.140,000

Working Capital Forecast = Total Current Assets – Total Current Liabilities

Working Capital Forecast = Rs.260,000 – Rs.140,000

Working Capital Forecast = Rs.120,000

2) Estimating the Cash Costs of Current Assets and Current Liabilities:

This method focuses on estimating the cash outflows related to current liabilities and the cash inflows related to current assets. It considers the timing of cash flows, providing insights into the actual cash requirements.

Example 22.2: XYZ Retail Company wants to forecast its working capital needs for the upcoming year. They estimate the following cash costs for current assets and current liabilities:

Cash Costs of Current Assets (Cash Outflows):

Cash Costs for Inventory: Rs. 500,000, Cash Costs for Prepaid Expenses: Rs. 20,000, Total Cash Costs of Current Assets: Rs. 520,000

Cash Costs of Current Liabilities (Cash Inflows):

Cash Payments for Accounts Payable: Rs.400,000, Cash Payments for Short-Term Debt: Rs.100,000, Total Cash Costs of Current Liabilities: Rs.500,000

Working Capital Forecast = Total Cash Costs of Current Assets – Total Cash Costs of Current Liabilities

Working Capital Forecast = Rs.520,000 – Rs.500,000

Working Capital Forecast = Rs.20,000

3) The Percent of Sales Method:

This approach estimates working capital needs as a percentage of sales revenue. It assumes that certain current assets and liabilities vary with sales volume.

Example 22.3: LMN Electronics Company wants to forecast its working capital requirements for the next year. Based on historical data, they find that their accounts receivable and inventory levels are typically 20% and 15% of annual sales, respectively. Their accounts payable level is typically 10% of annual sales.

Forecasted Sales for Next Year: Rs. 2,000,000

Working Capital Forecast = (Accounts Receivable + Inventory) – Accounts Payable

Working Capital Forecast = $(0.20 * \text{Rs. } 2,000,000) + (0.15 * \text{Rs. } 2,000,000) - (0.10 * \text{Rs. } 2,000,000)$

Working Capital Forecast = Rs. 400,000 + Rs. 300,000 – Rs. 200,000

Working Capital Forecast = Rs. 500,000

4) The Operating Cycle Approach:

This method estimates working capital needs based on the time it takes to convert resources into cash. It considers the time required for inventory conversion, receivables collection, and payables payment.

Example 22.4: PQR Manufacturing Company wants to forecast its working capital needs for the next six months. They estimate the following periods for their operating cycle components: Days Inventory Outstanding (DIO): 60 days

Days Sales Outstanding (DSO): 45 days

Days Payables Outstanding (DPO): 30 days

Working Capital Forecast = (DIO + DSO) – DPO

Working Capital Forecast = $(60 \text{ days} + 45 \text{ days}) - 30 \text{ days}$

Working Capital Forecast = 75 days – 30 days

Working Capital Forecast = 45 days

These methods provide different perspectives on estimating working capital needs. Businesses can use a combination of these approaches based on their specific requirements and available data to make more accurate working capital forecasts.

22.4.4 Sources of Working Capital

Working capital sources encompass the various means by which a company secures funds to meet its short-term obligations and support its day-to-day operations. These sources can be broadly categorised into short-term and long-term sources. Below are some common sources of working capital:

Short-Term Sources of Working Capital:

- a) **Trade Credit:** Companies often receive credit from suppliers, allowing them to procure goods or services and defer payment for a specified period, typically 30 to 90 days.
- b) **Bank Loans:** Short-term bank loans, such as working capital loans or overdraft facilities, offer quick access to funds to manage day-to-day operational expenses and cash flow needs.
- c) **Commercial Paper:** Creditworthy corporations may issue commercial paper, which are short-term unsecured promissory notes, to raise funds from the capital markets.
- d) **Factoring and Receivables Financing:** Companies can sell their accounts receivable to third-party factors at a discount, enabling them to obtain immediate cash against outstanding invoices.
- e) **Inventory Financing:** Inventory can be used as collateral to secure short-term loans or lines of credit.
- f) **Trade Advances:** Some businesses receive advance payments from customers for goods or services, providing an immediate infusion of working capital.

Long-Term Sources of Working Capital:

- a) **Equity Financing:** Companies can raise long-term working capital by issuing new shares of stock to investors through private placements or public offerings.
- b) **Debt Financing:** Long-term debt instruments such as corporate bonds or bank loans with longer repayment periods are used to raise working capital.
- c) **Retained Earnings:** Accumulated profits, known as retained earnings, can be reinvested in the business to serve as a long-term source of working capital.
- d) **Venture Capital and Private Equity:** Startups and high-growth firms may secure working capital from venture capital firms or private equity investors in exchange for ownership stakes.
- e) **Asset-Based Lending:** Companies can obtain long-term loans by using assets like real estate or equipment as collateral.

- f) **Debentures:** Long-term unsecured debt instruments called debentures can be issued to raise working capital from the capital markets.

22.4.5 Approaches for Determining the Financial Mix

Determining the right financing mix, or capital structure, is a crucial decision for any company, as it involves selecting the ideal proportion of debt and equity to finance its operations. The financing mix can have a significant impact on a company's risk profile, cost of capital, and overall financial performance. There are three common approaches to determining the financing mix: the hedging approach, the conservative approach, and the trade-off approach.

A) Hedging Approach:

The hedging approach aims to strike a balance between debt and equity financing to minimise the overall cost of capital while effectively managing financial risk. Companies adopting this approach calculate the optimal capital structure by minimising the weighted average cost of capital (WACC). WACC is the average cost of debt and equity, considering their respective proportions in the capital structure.

Companies following the hedging approach increase their debt levels as long as the cost of debt remains lower than the cost of equity. This strategy leads to a decrease in WACC, resulting in a lower overall cost of capital, which can potentially enhance the company's value.

However, the hedging approach recognises that excessive leverage can increase financial risk, especially if the company encounters challenges in meeting its debt obligations. Thus, there is a limit to how much debt a company should take on, and the optimal capital structure lies at the point where WACC is minimised without jeopardising the company's financial health.

B) Conservative Approach:

The conservative approach emphasises stability and financial security. Companies following this approach rely more on equity financing and maintain a lower level of debt in their capital structure. By having a higher proportion of equity, the company reduces financial risk and has a stronger ability to withstand economic downturns or unforeseen obstacles.

While the conservative approach provides a more secure financial position, it may result in a higher cost of capital compared to the hedging approach. Equity financing is often more expensive due to the costs associated with issuing new shares and the potential dilution of ownership for existing shareholders.

1) What are the methods of forecasting working capital?

.....

.....

.....

2) What are the sources of working capital?

.....

.....

.....

22.5 DIVIDEND POLICY

The establishment of a dividend policy is an important function of the finance manager. This section deals with the determination of dividend policy and the various forms of dividends.

The term dividend refers to that part of the profits of a company which is distributed amongst its shareholders. It may, therefore, be defined as the return that a shareholder gets from the company, out of its profits, on his shareholdings. According to the Institute of Chartered Accountants of India, a dividend is “a distribution to shareholders out of profits or reserves available for this purpose.”

A) Nature of Dividend Decision

The dividend decision of the firm is of crucial importance for the finance manager since it determines the amount of profit to be distributed among shareholders and the amount of profit to be required in the business (popularly termed as retained earnings) for financing its long-term growth. There is a reciprocal relationship between the cash dividends and retained earnings. Larger dividends result in less retained earnings. Fewer dividends result in larger retained earnings. While making dividend decisions, the management will consider the effect of the decision on the maximisation of shareholders' wealth. In case the payment of dividends helps the management in achieving this objective, it would be advisable to pay dividends. In case the payment of dividends does not help in achieving this objective, the management would be well advised to retain the profits and use them for financing investment programmes. Thus, the dividend decision is based on its impact on the value of the firm.

B) Conflicting Theories

There are conflicting theories regarding the impact of dividend decisions on the valuation of a firm. According to one school of thought, dividend decisions do not affect only the shareholders' wealth and so also the

valuation of the firm. According to another school of thought, dividend decision materially affects the shareholders' wealth and the valuation of the firm. For the sake of convenience of discussion, we can put the viewpoints of scholars under the following two groups:

1) Concept of Irrelevance of Dividend

This school of thought is associated with Soloman, Modigliani and Miller. According to them, dividend policy does not affect the share prices of a company and is, therefore, of no consequence. Investors do not differentiate between dividends and capital gains. Their basic desire is to earn a higher return on their investment. In case the company has adequate investment opportunities giving a higher rate of return than the cost of retained earnings, the investors would be content with the firm retaining the earnings. However, if the expected return on projects is likely to be less than what it would cost, the investors would prefer to receive the earnings (i.e. dividends). Thus, a dividend decision is a financing decision, i.e., whether to finance the company's funds requirements by retained earnings or not. In case the company has profitable investment opportunities, it will retain the earnings to finance them, otherwise distribute them. The shareholders are only interested in income whether it is in the form of dividends or capital gains.

2) Modigliani and Miller's Approach

Franco Modigliani and Merton Miller have expressed their opinion more comprehensively. They have opined that the price of shares of a firm is determined by its earning potentiality and investment policy and never by the pattern of income distribution. As observed by them, "under conditions of perfect capital markets, rational investors, absence of tax discrimination between dividend income and capital appreciation, given the firm's investment policy its dividend policy may not influence the market price of the shares."

The logic put forward by Modigliani and Miller in support of their hypothesis is that whatever increase in shareholders' wealth results from dividend payments, will be exactly offset by the effect of raising additional capital. For example, if a company, having investment opportunities, distributes all its earnings among the shareholders, it will have to raise the capital required from outside. This will increase the number of shares, resulting in a fall in the future earnings per share. Thus, whatever a shareholder has gained because of increased dividends will be neutralised completely on account of a fall in the value of shares due to a decline in the expected earnings per share.

22.6 CAPITAL BUDGETING

Capital budgeting is the process of long-term planning, encompassing proposed capital expenditures and their funding. This comprehensive approach involves both raising long-term funds and allocating them appropriately. It refers to the formal procedure a company employs to acquire

and invest in capital. Through capital budgeting, firms evaluate major fixed asset purchases, including decisions about adding, disposing, modifying, or replacing such assets.

It is essential to note that capital budgeting decisions also extend to investments in current assets that arise due to fixed asset investments. For instance, when implementing a new distribution system, the decision to invest in a new warehouse and additional inventory must be evaluated as a single package, not as separate investments.

Cases of capital budgeting decisions

Capital budgeting decisions are prevalent in various industries and organisations. Below are some cases that illustrate several types of capital budgeting decisions:

- 1) **New Product Development:** A manufacturing company is considering the launch of a new product line. The company needs to evaluate the feasibility of the project by estimating the initial investment required for production facilities, marketing costs, and expected cash flows over the next few years. They will use capital budgeting techniques like NPV, IRR, and Payback Period to assess whether the investment is financially viable and aligned with the company's strategic objectives.
- 2) **Expansion of Manufacturing Facilities:** A successful retail chain wants to expand its operations by opening new stores in different regions. The company must decide how much to invest in new store construction, inventory, and staffing. Capital budgeting analysis will help them determine the expected returns and potential risks associated with each location, allowing them to allocate resources to the most profitable expansion opportunities.
- 3) **Replacement of Machinery:** An industrial plant has ageing machinery that requires frequent maintenance, leading to increased downtime and inefficiencies. The management needs to evaluate whether investing in new, more modern equipment would improve productivity and reduce operational costs. By comparing the costs of new machinery with the expected cost savings and increased cash flows, they can make an informed capital budgeting decision.
- 4) **Mergers and Acquisitions:** A technology company is considering acquiring a smaller firm with innovative products that complement its existing offerings. Before making the acquisition, the acquirer must conduct a thorough capital budgeting analysis to determine the potential synergies, expected cash flows, and the impact of the acquisition on shareholder value.
- 5) **Research and Development (R&D) Projects:** A pharmaceutical company is deciding whether to invest in the development of a new drug. The R&D project will involve significant upfront costs and take several

years before the product is commercialised. Capital budgeting techniques, along with scenario analysis, will help assess the probability of success, potential market demand, and the overall financial viability of the R&D project.

- 6) **Capital Improvement Projects:** A municipal government needs to decide whether to invest in upgrading infrastructure, such as bridges, roads, or water treatment plants. Capital budgeting analysis will consider the initial investment, the estimated life of the improvement, maintenance costs, and the benefits to the community, including potential economic growth and increased property values.

22.6.1 Importance of Capital Budgeting

Capital budgeting holds immense significance for businesses and organisations due to various compelling reasons:

- 1) **Long-term investment choices:** Capital budgeting involves making decisions about long-term investments that have enduring effects. These investments typically require substantial financial resources, and their success or failure can profoundly impact the company's future growth and profitability.
- 2) **Optimal resource allocation:** By utilising capital budgeting techniques, companies can efficiently allocate their limited resources, such as funds and workforce, to projects that are expected to yield the highest returns. This ensures that resources are utilised effectively to create maximum value for the organisation.
- 3) **Maximising shareholder wealth:** The main goal of capital budgeting is to pinpoint and choose projects that contribute to the growth of the company's shareholders' wealth. By investing in projects with favourable net present values (NPVs) or significant internal rates of return (IRRs), the company can boost shareholder value and appeal to potential investors.
- 4) **Strategic alignment:** Capital budgeting decisions are closely tied to a company's strategic objectives. It ensures that projects chosen for investment align with the organisation's long-term goals, enabling the allocation of resources to endeavours that contribute to the company's overall vision.
- 5) **Risk assessment:** Capital budgeting involves a comprehensive evaluation of the risks associated with different investment projects. Understanding potential risks and uncertainties enables companies to make informed decisions and implement risk mitigation strategies when necessary.
- 6) **Capital rationing:** In situations where capital is limited, capital budgeting helps prioritise investment opportunities based on their

expected returns and strategic importance. It enables companies to optimise the use of available funds by selecting the most promising projects.

- 7) **Efficient use of funds:** Capital budgeting prevents investments in projects with negative NPVs or low returns, thus avoiding wasteful use of financial resources. Focusing on profitable projects allows companies to utilise their funds efficiently to achieve growth and success.

22.6.2 Kinds of Capital Investment Proposals

A firm may be presented with multiple investment proposals, and their decision to adopt one, some, or all of them depends on their categorisation as independent, contingent, or mutually exclusive.

a) Independent Proposals:

These investment proposals are non-competing, implying that accepting one proposal does not prevent the acceptance of another. The firm can directly assess each proposal's return on investment and accept those that offer returns higher than the desired rate of return.

Proposals meeting this criterion are accepted, while the rest are rejected.

b) Contingent or Dependent Proposals:

These proposals' acceptance relies on the approval of one or more other proposals. For instance, the acquisition of a new machine may be contingent upon a substantial plant expansion. In such cases, the dependent investment proposal should be presented along with the related proposal it relies on, providing a comprehensive perspective.

c) Mutually Exclusive Proposals:

These proposals are competing in such a way that accepting one automatically excludes the acceptance of others. **For instance**, if a company is evaluating two temperature control systems and decides to invest in one system, it means rejecting the other. Consequently, the firm needs to employ a decision-making technique to choose the most favourable option, leading to the exclusion of other alternatives.

Check Your Progress 4

- 1) What are the theories regarding the impact of dividend policy?

.....

.....

.....

- 2) What is capital budgeting? Discuss what is its importance?

.....
.....
.....

22.6.3 Factors Affecting Capital Investment Decision

Factors affecting capital investment decisions can vary depending on the industry, company, and specific project. Below are some common factors that influence such decisions:

- 1) **Market Demand:** Market demand for the product or service being offered is a fundamental factor influencing capital investment decisions. Companies need to assess the potential size of the target market, the growth rate, and the demand forecast to determine the potential sales volume and revenue generation. **Example:** Regarding the smartphone project, a comprehensive market analysis suggests that the smartphone market is projected to experience an annual growth rate of 8%, and the company's product is anticipated to secure a 10% market share.
- 2) **Competition:** The level of competition in the market is a critical consideration. A highly competitive market may lead to price wars and reduced profit margins, affecting the project's financial feasibility. **Example:** The smartphone market has several well-established brands with strong customer loyalty, making it a highly competitive space for new entrants.
- 3) **Cost of Capital:** The cost of capital signifies the return that investors anticipate for funding an investment. It plays a crucial role in establishing the discount rate utilised to assess future cash flows. **Example:** If the company's cost of capital is 12%, it implies that any investment must generate returns exceeding 12% to be deemed financially feasible.
- 4) **Initial Investment Cost:** The total upfront cost required to start the project, including machinery, equipment, land, and working capital, needs to be carefully evaluated. **Example:** The initial investment for the smartphone project includes ₹7,00,00,000 for machinery, ₹2,00,00,000 for land, and ₹1,00,00,000 for working capital.
- 5) **Cash Flows:** Projected cash inflows and outflows over the life of the project are essential for evaluating profitability. Companies must estimate expected revenue and operating expenses. **Example:** The smartphone project expects ₹20,00,00,000 in annual sales revenue and annual operating expenses of ₹12,00,00,000.
- 6) **Payback Period:** The payback period is a measure of the time required to recoup the initial investment. Typically, a shorter payback period is

considered more favourable. **Example:** In the case of the smartphone project, with a payback period of 0.5 years, it means that the initial investment will be recovered within six months.

- 7) **Net Present Value (NPV):** NPV (Net Present Value) evaluates the present value of future cash flows minus the initial investment. A positive NPV indicates that the project is expected to generate more cash than the initial investment. **Example:** In the case of the smartphone project, with an NPV of approximately ₹3,18,72,197, it suggests that the project is financially appealing.
- 8) **Internal Rate of Return (IRR):** IRR (Internal Rate of Return) is the discount rate at which the NPV of the project becomes zero. It represents the project's rate of return. **Example:** in the given example, an IRR of approximately 18%, signifies the expected return on the investment.

22.6.4 Computation of Capital Investment

The computation of capital investment required involves calculating the total amount of funds needed to initiate and implement a specific project or investment opportunity. It encompasses various components and expenses that contribute to the overall cost of the project. The capital investment required can be calculated using the following steps:

Step 1: Identify Project Components:

When computing the capital investment required, the first step is to identify all the components and expenses that are essential for the successful initiation and implementation of the project. These components could include:

- 1) **Fixed Assets:** This involves the cost of acquiring long-term assets such as machinery, equipment, land, buildings, vehicles, and any other infrastructure needed for the project.
- 2) **Working Capital:** It refers to the funds required to finance the day-to-day operations of the project, including inventory, raw materials, and operational expenses.
- 3) **Initial Setup Costs:** This category includes expenses related to project planning, feasibility studies, legal fees, permits, licenses, and any other preliminary costs necessary to get the project off the ground.
- 4) **Research and Development (R&D):** For projects that involve product development or innovation, R&D expenses must be accounted for.
- 5) **Marketing and Promotion:** For projects that require market entry or product promotion, marketing and promotional expenses need to be considered.

Step 2: Determine Costs:

After identifying the project components, the next step is to determine the costs associated with each of them. Companies can use various methods to ascertain these costs, including obtaining quotations from suppliers, conducting market research, or relying on historical data.

Example 22.5: Let us assume the following costs for a manufacturing project:

Fixed Assets: Machinery – ₹5,00,000, Equipment – ₹3,00,000, Land – ₹10,00,000, Building – ₹20,00,000, Working Capital: ₹8,00,000, Initial Setup Costs: ₹2,00,000, Research and Development: ₹1,50,000, Marketing and Promotion: ₹1,00,000

Step 3: Sum the Costs:

Once the costs for each component are determined, the next step is to add up all the costs to get the total capital investment required.

Example 22.6: Total Capital Investment Required = ₹5,00,000 + ₹3,00,000 + ₹10,00,000 + ₹20,00,000 + ₹8,00,000 + ₹2,00,000 + ₹1,50,000 + ₹1,00,000 = ₹50,50,000

In this example, the total capital investment required for the project is ₹50,50,000.

Step 4: Consider Contingencies and Contingent Expenses:

To account for unforeseen expenses, potential changes in costs, or unexpected events, it is advisable to include a contingency buffer.

Example 22.7: Let us assume a contingency of 10% of the total costs: Contingency (10% of total costs) = $0.10 * ₹50,50,000 = ₹5,05,000$

Step 5: Final Capital Investment Required:

Finally, add the contingency amount to the total costs to get the final capital investment required. **Example 22.8:** Final Capital Investment Required = ₹50,50,000 + ₹5,05,000 = ₹55,55,000

In this example, the final capital investment required for the project, including the contingency, is ₹55,55,000.

It is important to note that the capital investment required should be carefully analysed and compared with the expected returns and potential risks associated with the project. This comprehensive evaluation helps companies make well-informed decisions regarding their capital investments and ensures the efficient allocation of resources to maximise the project's success and overall company growth.

22.6.5 Forecasting Future Costs

Forecasting future costs is a crucial aspect of capital budgeting as it enables businesses to estimate the financial implications of an investment project accurately. When making capital budgeting decisions, organisations need to project the costs associated with the proposed project over its entire life cycle. Accurate cost forecasting helps in evaluating the feasibility, profitability, and risk of the investment, allowing management to make well-informed decisions. Here are some steps and considerations for forecasting future costs during capital budgeting:

- 1) **Comprehensive Project Analysis:** Start by conducting a thorough analysis of the proposed project. Understand its scope, objectives, and the resources required for successful implementation. Consider all cost elements, including initial investment, operating expenses, maintenance costs, and potential expansion or replacement costs.
- 2) **Historical Data and Benchmarking:** Use historical data from similar projects or industry benchmarks to gain insights into potential cost patterns. This data can help in making more accurate assumptions and estimates for future costs.
- 3) **Inflation and Price Fluctuations:** Account for inflation and potential price fluctuations of resources, materials, and labour over the project's duration. These factors can significantly impact project costs over time.
- 4) **Economic and Market Conditions:** Consider the current economic conditions and future market trends that could influence the cost of inputs, labour, and other resources.
- 5) **Consultation with Experts:** Engage subject matter experts and professionals who have experience in similar projects. Their insights and knowledge can provide valuable inputs for cost estimation.
- 6) **Sensitivity Analysis:** Perform sensitivity analysis by evaluating the potential impact of changes in cost assumptions on the project's financial viability. This helps identify the most critical cost drivers and the level of uncertainty in cost projections.
- 7) **Cost Breakdown Structure:** Develop a cost breakdown structure that categorises and itemises various cost components, such as direct costs, indirect costs, fixed costs, variable costs, and overheads.
- 8) **Contingency Planning:** Incorporate contingency allowances in the cost projections to account for unforeseen events and potential cost overruns.
- 9) **Cash Flow Projections:** Prepare cash flow projections that outline the timing and magnitude of costs throughout the project's life cycle. This aids in assessing the project's impact on the company's cash flow.

Check Your Progress 5

- 1) What are the factors affecting capital investments?

.....

.....

.....

- 2) Define the process of computation of capital investment.

.....

.....

.....

22.7 LET US SUM UP

The present unit discusses about Corporate Finance. Corporate finance is the area of finance that deals with the sources of funding, and the capital structure of corporations, the actions that managers take to increase the value of the firm to the shareholders, and the tools and analysis used to allocate financial resources. The primary goal of corporate finance is to maximise or increase shareholder value. There are two sources of funds for short term, medium-term and long-term needs of a company. These sources are Internal (depreciation and retained earnings) and external (loans, equities and bonds or debentures). We also discussed about Capital Structure. It refers to the composition and proportion of various sources of funds that a company employs to finance its operations, investments, and other financial activities. Components of Capital Structure are: **Owner's capital** and **Borrowed Capital**. Another concept which is crucial to understand in financial management is **Working Capital**. It is calculated by subtracting current liabilities from current assets. Working capital is determined by: nature of business, business cycle and seasonality, sales volume and growth rate, credit Policy etc. We also discussed the approaches for determining the financial mix: the hedging approach, the conservative approach, and the trade-off approach. We also discussed about Capital Budgeting and factors affecting capital investment decisions. Finally, we discussed about **computation of capital investment**. The computation of capital investment required involves calculating the total amount of funds needed to initiate and implement a specific project or investment opportunity.

22.8 KEY WORDS

- Managing Money** : Managing money is the process of overseeing and planning your income and expenses to achieve your financial goals.
- Capital Structure** : Capital structure refers to the combination of debt and equity that a company uses to finance its

operations and growth. It is the composition of long-term funds deployed by a firm.

Capital Budgeting : Capital budgeting refers to the process that companies use to make investment decisions in long-term projects and capital assets. It involves allocating capital funds to projects expected to generate profits for the company over a period.

Investment proposals : An investment proposal is a document that outlines the details of a potential investment opportunity to attract investors or funding. It serves as a formal and structured way to present key information about a business, project, or venture to convince potential investors to commit their capital.

22.9 SUGGESTED READINGS

- Graham, Benjamin. 2003. The Intelligent Investor. New York, NY: Harper Business.
- Damodaran, A. (2011). The little book of valuation: How to value a company, pick a stock and profit. John Wiley & Sons.

22.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Preference Shares, Equity Shares
- 2) Source of internal finance through its non-cash item in books, but it is a recognition of the value of the asset which will reduce each year and attempting to identify the amount of cash to set aside when the asset is no longer in use to us.
- 3) Resembling an alternative to loans, secured debentures require collateral for fund generation. Requiring no collateral, unsecured debentures are less favoured during payments compared to secured debentures.

Check Your Progress 2

- 1) Owner's capital refers to the portion of a company's assets that belongs to its owners or shareholders. Borrowed capital, also known as debt capital, refers to funds that a company raises by borrowing money from external sources, such as financial institutions, investors, or bondholders.
- 2) Business risk, tax considerations, cost of capital, market conditions, company size and life cycle, flexibility and control, asset structure,

lender and investor preferences, regulatory environment, market perception.

Check Your Progress 3

- 1) Refer to Sub-section 22.4.3
- 2) Trade Credit, Bank Loans, Commercial Paper, Factoring and Receivables Financing, Inventory Financing, Trade Advances, Equity Financing, Debt Financing, Retained Earnings, Venture Capital and Private Equity, Asset-Based Lending, Debentures.

Check Your Progress 4

- 1) Concept of Irrelevance of Dividend, Modigliani and Miller's Approach.
- 2) Capital budgeting is the process of long-term planning, encompassing proposed capital expenditures and their funding. It helps in long-term investment choices, optimal resource allocation, maximising shareholder wealth, Strategic alignment, Risk assessment, Capital rationing.

Check Your Progress 5

- 1) Market demand, competition, cost of capital, initial investment cost, cash flows, payback period, net present value (NPV), internal rate of return (IRR)
- 2) Refer to Sub-section 22.6.4

UNIT 23 FOREIGN DIRECT INVESTMENT AND FOREIGN PORTFOLIO INVESTMENT

Structure

- 23.0 Objectives
- 23.1 Introduction
- 23.2 Concept of FDI
- 23.3 Methods of FDI
- 23.4 Types of FDI
- 23.5 Routes of FDI
- 23.6 Advantages and Limitations of FDI
- 23.7 Highlights of FDI Policy, 2020
- 23.8 Concept of FPI
- 23.9 Difference between FDI and FPI
- 23.10 Categories of FPI
- 23.11 Advantages and Disadvantages of FPI
- 23.12 Eligibility Criteria of FPI in India
- 23.13 Let Us Sum Up
- 23.14 Key Words
- 23.15 Some Useful Books
- 23.16 Answers/Hints to Check Your Progress Exercises

23.0 OBJECTIVES

After going through this unit, you will be able to:

- explain the concept of FDI;
- describe the various methods and types of FDI;
- discuss the various benefits and limitations of FDI;
- state the multiple routes and regulations governing FDI;
- explain the concept of FPI;
- differentiate between FDI and FPI;
- state the various benefits of FPI; and
- explain the provisions concerning FPI.

23.1 INTRODUCTION

Developing countries like India are attractive destinations for Foreign Direct Investment (FDI). FDI is an important source of external private finance. FDI is a way of integrating international economics as it creates and promotes stable and long-lasting links between economies. It offers the following major benefits to economies: a) promotes international trade through access to foreign markets, b) facilitates the technology transfer between countries, c) leads to economic development, and d) creates job opportunities. During the financial year 2021-22, India received foreign direct investment inflows of USD 82.274 billion and USD 54.934 billion in the first three quarters of the financial year 2022-23. Various initiatives like policy reforms, relaxation in regulations etc. are being taken to boost foreign investment.

Similarly, Foreign Portfolio Investment (FPI) is essential in modern economies with a sophisticated financial system. FPI allows foreign parties to invest in financial assets and instruments in the host country. This has a positive bearing on the price of domestic assets.

23.2 CONCEPT OF FDI

Foreign direct investment (FDI) can be defined as an investment made outside the home country where an investor resident in one country invests in another country in a way that establishes a long interest in and a high degree of influence over an enterprise. Through such investment, the investor owns 10 per cent or more of the voting power and thereby can exercise a high degree of influence in the management and operations of the foreign business.

Definition of FDI

The International Monetary Fund (IMF) defines FDI as: “A category of international investment that reflects the objective of a resident entity in one economy (direct investor or parent enterprise) obtaining a lasting interest and control in an enterprise resident in another economy (direct investment enterprise)”.

FDI or ‘Foreign Direct Investment’ means investment through capital instruments by a person resident outside India in an unlisted Indian company, or in ten per cent or more of the post-issue paid-up equity capital on a fully diluted basis of a listed Indian company.

23.3 METHODS OF FDI

The most common methods of Foreign Direct Investment (FDI) are:

- 1) **Greenfield Investment:** It refers to a type of investment where a foreign entity like a company, an individual, or an organization, invests in

another country by establishing a new business or by expanding its existing operations. For example, in the year 2019 Samsung which is a South Korean company invested \$500 million in India to set up a mobile phone manufacturing plant in Noida, Uttar Pradesh, for manufacturing smartphones and other electronic devices.

- 2) **Mergers and Acquisitions (M&A):** It refers to investing (FDI) for acquiring or merging with an existing foreign company. For example, in the year 2017 Vodafone which is a British Telecommunications Company merged its Indian operations with Idea Cellular and thus created one of the largest Telecom service providers in India.
- 3) **Joint Ventures:** Foreign Direct Investment (FDI) in joint ventures is a popular business in India. It allows foreign companies to collaborate and invest in various sectors of the Indian economy. For example, General Electric (GE) formed a joint venture with the Indian conglomerate Tata Group in 2015 to create Tata Consultancy Services (TCS), a business process outsourcing company.
- 4) **Subsidiary Companies:** FDI in subsidiary companies results in establishing a wholly-owned subsidiary in a foreign country. For example, Suzuki Motor Corporation owns a majority stake in Maruti Suzuki India Ltd., a subsidiary that manufactures and sells cars in India.

23.4 TYPES OF FDI

The various types of FDI are:

- 1) **Horizontal FDI:** Investing funds in a company other than the domestic company that belongs to the same industry as that owned or operated by the FDI investor is called horizontal FDI. In other words, when a company initiates a similar operation or business model in another country, it refers to Horizontal FDI. For example, suppose a foreign company manufacturing apparel invests in or purchases an Indian company dealing in the same line of business which is apparel. Both companies belong to the same industry which is apparel, such FDI is therefore an example of horizontal FDI.
- 2) **Vertical FDI:** This form of FDI refers to the investments that are made within a supply chain in a company, which may or may not belong to the same industry. It occurs when a multinational acquires an operation that either acts as a supplier or distributor. It can be said that when vertical FDI happens, a business invests in an overseas firm which may supply or sell products.

Vertical FDIs can take two ways backward vertical integration and forward vertical integration. For example, A foreign fast-food chain invests in agricultural operations in India to secure a steady supply of fresh produce like vegetables and meat for its restaurant outlets.

Forward integration in FDI involves a foreign company expanding its operations downstream in the supply chain. This means getting involved in distribution, marketing, or retail activities. For example, a foreign pharmaceutical company opens its own pharmacies or healthcare clinics in India to sell its medicines directly to patients, bypassing traditional distribution channels.

- 3) **Conglomerate FDI:** Such a form of FDI occurs when a company invests in a company other than the domestic company that is a counterpart of the investing company. The company that is invested in belongs to an altogether different industry. Such investments help in creating more niches or to begin a new business in another country. For example, General Electric has engaged in conglomerate FDI over the years. The company has invested in various industries like aviation, healthcare, renewable energy, and financial services.
- 4) **Platform FDI:** It refers to investments made by multinational corporations in a specific location referred to platform to serve as a base for conducting different business activities and expanding their operations both regionally and globally. These platforms act as hubs for manufacturing, services, research and development, and more. Many global IT giants, including IBM, Microsoft, and Accenture, have set up extensive operations in India. These companies use India as a platform for software development, IT consulting, and application maintenance services, serving clients worldwide.

Check Your Progress 1

- 1) Define Foreign Direct Investment.

.....

.....

.....

.....

- 2) Examine the different methods of Foreign Direct Investment (FDI).

.....

.....

.....

.....

- 3) Explain the different types of Foreign Direct Investment.

.....

.....

.....

.....

23.5 ROUTES OF FDI

In this section, we discuss the various routes through which FDI arrives in the host country concerning the Indian Economy. FDI offers lots of opportunities and contributes significantly to the economic development of the country. Keeping this in mind, the Indian government has been taking steps to promote foreign investment in India like relaxation in regulatory compliances, streamlining the complexities involved in the investment procedures etc. The FDI policy specifies the regulations governing the FDI, the areas where the FDI is allowed, the areas where the FDI is restricted, routes of FDI etc. FDI can be made through Automatic Routes and Government Routes.

Automatic Route

FDI through Automatic Route does not require investors to seek approval from the Government of India for the investment. Foreign investment in sectors like agriculture and animal husbandry, plantation, mining, petroleum and natural gas, airports, construction, manufacturing, industrial parks, e-commerce activities, railway infrastructure, asset construction companies, credit information companies, and pharmaceuticals are allowed under the automatic route.

Government Route

FDI through the Government Route requires investors to seek prior approval from the Government of India. Proposals for foreign investment under the Government Route are considered by the respective Administrative Ministry/Department. Foreign investment in sectors like mining, defence, broadcasting, print media/digital media, civil aviation, satellites, telecommunications, and private security agencies are the areas where prior approval from the Indian government is required and regulations and procedures as prescribed under FEMA (defined hereinafter) and by RBI must be followed.

However, there are certain areas where FDI is strictly prohibited under any of the above-mentioned routes due to reasons like national security, protection of the nation's interest etc. These areas include the lottery business, gambling, and betting, chit funds, Hundis and Nidhi companies, trading in transferable development rights, real estate business, manufacturing of cigars, cheroots, cigarillos and cigarettes, of tobacco or tobacco substitutes. The sectors that are not open to private sector investment include atomic energy and railway operations.

23.6 ADVANTAGES AND DISADVANTAGES OF FDI

FDI offers several benefits. These are:

- 1) **Contribution to Economic Growth:** FDI contributes significantly to the economic growth of the host country by injecting capital into various sectors that lead to increased production, consumption, and overall economic activity.
- 2) **Jobs creation:** FDI creates new jobs in the host country as foreign companies establish or expand their operations, hiring local workers and professionals.
- 3) **Bringing Technology:** Foreign investors bring advanced technology, expertise, and management practices that increase the productivity and competitiveness of the local industries.
- 4) **Increase in exports:** This leads to an increase in exports from the host country as foreign companies may use local resources and labour to produce goods and services for global markets.
- 5) **Infrastructure development:** To attract foreign investments in the country, host countries develop their infrastructure by investing in transportation, energy, and telecommunications. Such development in infrastructure benefits the entire economy.
- 6) **Setting up of new industries:** FDI helps in diversifying the host country's economy by setting up new industries and developing new sectors.
- 7) **Access to new markets:** Local firms can benefit from foreign investors' global networks, gaining access to new markets and distribution channels.
- 8) **Increased government revenue:** FDI also results in enhancing government revenue by way of taxes.

Disadvantages of FDI:

The following are certain disadvantages of FDI:

- 1) FDI may result in a loss of control over their domestic industries which can also affect the ability to make independent decisions.
- 2) Due to the policies of foreign investors, a country may become vulnerable to economic fluctuations leading to economic instability.
- 3) Foreign investors often take profits back to their home countries, reducing the amount of money reinvested in the host country's economy.
- 4) FDI even results in environmental degradation as foreign investors may prioritize profit over environmental sustainability.
- 5) Host countries may face challenges in protecting their intellectual property rights when collaborating with foreign investors, potentially leading to knowledge and technology leakage.

- 6) Increased FDI may sometimes intensify competition for local resources, potentially driving up costs and pricing local businesses out of the market.

23.7 HIGHLIGHTS OF FDI POLICY 2020

FDI is regulated primarily by the Department of Promotion of Industry and International Trade (DPIIT), under the Government of India. The latest policy released by DPIIT is the Consolidated Foreign Direct Investment (FDI) Policy of 2020, effective from October 15, 2020. Major highlights of this policy are discussed in this section. The FDI policy of a country is revised as per the changing needs and economic environment of the country. Some of the highlights of FDI policy 2020 are:

1) List of Eligible investors:

Non-resident entities can invest in India under the FDI Policy, with certain sectors/activities prohibited. NRIs residing in Nepal and Bhutan, as well as citizens of these countries, can invest in Indian companies on a repatriation basis. Overseas Corporate Bodies (OCBs) not under RBI scrutiny can invest as non-resident entities under the FDI Policy and Foreign Exchange Management (Non-Debt Instrument) Rules, 2019. Companies, trusts, and partnership firms incorporated abroad and controlled by NRIs can invest in India under special dispensation. Foreign Portfolio Investors (FPIs) and Foreign Venture Capital Investors (FVCIs) are also permitted to make investments, with registered FPIs and NRIs able to trade through recognized Indian Stock Exchanges via registered brokers.

2) Eligible Investee Entities:

It includes Indian companies authorized to issue capital for FDI. NRIs can invest in partnership firms or proprietary concerns on a non-repatriation basis, using inward remittances or funds from specified accounts, provided the businesses are not in agriculture, plantation, real estate, or print media. Trusts cannot receive foreign investments except for SEBI-regulated VCFs. LLPs can receive foreign investment under the automatic route, allowing downstream investments and conversion into companies where permitted. Investment vehicles and startups can also attract foreign investments under specific guidelines outlined in the Foreign Exchange Management Rules, 2019.

3) Entry Routes for Investment:

Non-residents can invest in Indian companies' equity shares, convertible debentures, or preference shares through either the Automatic Route or the Government Route.

Prohibited sectors for FDI include:

- Lottery business (government/private, online, etc.)
- Gambling and betting activities, including casinos
- Chit funds
- Nidhi companies
- Trading in Transferable Development Rights (TDRs)
- Real estate business or construction of farmhouses (excluding specific exceptions)
- Manufacturing of cigars, cheroots, tobacco etc.
- Certain activities/sectors closed to private sector investment include Atomic Energy and railway operations, except for specific permitted activities. Additionally, foreign technology collaboration in any form, such as licensing for franchises, trademarks, brand names, or management contracts, is prohibited for lottery business, gambling, and betting activities.

Percentage of FDI Allowed in Major Sectors:

- Agriculture & Animal Husbandry: 100% (through automatic route)
- Plantation Sector: 100% (through automatic route)
- Mining: 100% (through automatic route)
- Mining and mineral separation of titanium: 100% (through government route)
- Petroleum & Natural Gas: 100% (through automatic route)
- Petroleum refining by the Public Sector Undertakings (PSU): 49% (through automatic route)
- Defence Manufacturing: 100% (Automatic up to 49% Above 49% under Government route in cases resulting in access to modern technology in the country)
- E-commerce activities: 100% (through automatic route)
- Multi Brand Retail Trading: 51% (through government route)
- Insurance: 49% (through automatic route)
- Banking- Private Sector: 74% (Automatic up to 49% above 49% & up to 74% under Government route)

- Banking- Public Sector: 20% (through government route)
- Food products manufactured or produced in India: 100% (through government route).

For more details on FDI Policy, 2020 refer https://dpiit.gov.in/sites/default/files/FDI-PolicyCircular-2020-29October2020_0.pdf.

Check Your Progress 2

- 1) Elaborate on the different pathways (routes) for Foreign Direct Investment (FDI).

.....

.....

.....

.....

- 2) Discuss the various benefits of FDI.

.....

.....

.....

.....

- 3) What are the disadvantages of FDI?

.....

.....

.....

.....

23.8 CONCEPT OF FOREIGN PORTFOLIO INVESTMENT (FPI)

In the Indian context, FPI has been defined as “Foreign portfolio investment” means any investment made by a person resident outside India through equity instruments where such investment is less than 10% of the post-issue paid-up share capital on a fully diluted basis of a listed Indian company or less than 10% of the paid-up value of each series of an equity instrument of a listed Indian company *Rule 2(t) of FEM (Non-debt Instruments) Rules, 2019 and para 2.1.20 of FDI Policy dated 15-10-2020.*

Foreign Portfolio Investment refers to investments made by foreign individuals, institutions, or funds into financial assets in another country like stocks, bonds, or other securities.

Securities and Exchange Board of India is the regulatory body that regulates FPI in India. The National Securities Depository Limited (NSDL) maintains a list of all the registered FPIs in India and this list of registered FPIs is publicly available on their website.

23.9 DIFFERENCE BETWEEN FDI AND FPI

FDI and FPI refer to investments that involve investing in foreign countries. Foreign Direct Investment refers to investments that require a long-term commitment that leads to the establishment of business interest in the foreign country whereas FPI refers to investments that are short-term in nature and that are intended to diversify investment portfolios. Such investments also allow the investors to take part in the growth of foreign economies. The key differences between FDI and FPI are:

- In FDI a direct business interest is established overseas whereas FPI is not a direct investment.
- An FDI leads to the sharing of resources, transfer of technology know-how, and funds and involves a joint venture or setting up of a subsidiary. IN FPI no such transfer of knowledge and technology takes place.
- Foreign direct investment is for a longer duration than foreign portfolio investment. FDI may include establishing a manufacturing unit, acquiring a company, or entering into a joint venture. FPI is a short-term form of investment and is done to diversify investment portfolios and take benefit of short-term market opportunities.
- FDI is made by institutions or venture capital companies whereas FPI simply involves investing in securities or assets outside the country.
- FPI is highly liquid and can be purchased and sold easily. Whereas FDI is not that liquid as it is a long-term investment.
- FPI involves passive investors, and FDI is all about active investors.
- In FDI, the investor enjoys a say in the decision-making and actively participates in the operations and administration of the company. However, in the case of FPI, the investor does not have any say in the management of the company.
- The risk in FDI is higher than in FPI. FDI is subject to various risks that relate to the political situation, economic conditions regulatory provisions and structure risks in the host country. Whereas FPI is subject to the risk that is associated with market volatility and is highly affected by fluctuations and changes in the currency, interest rates, etc.

23.10 CATEGORIES OF FPI

FPI is typically categorized under three categories as explained under:

- Category I: This category is regarded as a low-risk category. It comprises government/government-related establishments like central banks and international agencies like the World Bank, and IMF. For example, a sovereign wealth fund is a Category I FPI.
- Category II: This category involves a medium degree of risk. This category comprises mutual funds, insurance firms, asset management companies, banks, and pension funds.
- Category III: This category is the riskiest among all. It includes all other forms of investments that do not fall into Category I and Category II
Example: Private equity funds.

Check Your Progress 3

- 1) What do you understand by FPI?

.....

.....

.....

.....

- 2) Differentiate between Foreign Direct Investments and Foreign Portfolio Investments.

.....

.....

.....

.....

- 3) Describe the various categories of Foreign Portfolio Investment.

.....

.....

.....

.....

23.11 ADVANTAGES AND DISADVANTAGES OF FPI

The following are the advantages offered by FPI:

- 1) Foreign portfolio investments stimulate demand for companies' stocks, facilitating the raising of capital in a cost-effective manner.
- 2) It offers greater diversification to investors which lead to extending potential benefits from this broader investment base.

- 3) Investors can also benefit from fluctuations in exchange rates.
- 4) Overseas markets offer investors access to larger markets that may sometimes have less competition compared to their local market. This implies they can enjoy reduced competition in foreign countries.
- 5) FPI also offers liquidity ensuring that investors have the agility to capitalize swiftly on favourable opportunities.
- 6) FPI can boost stock market performance by increasing demand for shares of Indian companies. This, in turn, can benefit both companies and individual investors.
- 7) FPI brings foreign capital into the Indian financial markets that can be used for various purposes like infrastructure development, job creation, and economic growth.

Disadvantages of FPI

The various disadvantages associated with Foreign Portfolio Investment (FPI) are:

- 1) Large inflows and outflows of foreign capital in the form of FPI can destabilize financial markets and exchange rates. In times of uncertainty like economic slowdown, foreign investors withdraw their investments which may lead to capital flight.
- 2) It may sometimes lead to speculative behaviour in financial markets, as short-term investors seek quick profits. Such speculation can create market bubbles and lead to market volatility.
- 3) FPI exposes a country's currency to exchange rate fluctuations which may result in risks for both investors and the country's economy.
- 4) Foreign investors in FPI typically do not have a long-term commitment to the country and therefore such foreign investors may not align with the country's long-term economic goals.
- 5) FPI can encourage herd behaviour among investors which may lead to a cascading effect that can cause market panics.
- 6) FPI prioritizes short-term profits over the long-term interests of local businesses. This results in creating pressure on companies to focus on quarterly results at the expense of long-term growth and sustainability.
- 7) FPI also attracts manipulative activities by foreign investors who exploit regulatory gaps or engage in unethical trading practices.
- 8) Managing and regulating FPI flows can be challenging for governments, requiring strong oversight and regulatory frameworks to prevent abuses and maintain financial stability.

23.12 ELIGIBILITY CRITERIA FOR FOREIGN PORTFOLIO INVESTMENT IN INDIA

In this short section, we spell out some of the eligibility criteria to be considered a foreign portfolio investor. An individual must fulfil the following conditions to register as Foreign Portfolio Investor:

- The applicant must be a non-resident Indian as per the Income Tax Act 1961.
- The applicant must not be a citizen of a country that has been categorised under the public statement of FATF.
- The applicant is eligible to invest in securities of other countries.
- For investing in securities, the applicant must be authorized to do so by Memorandum of Association and Articles of Association Agreement.
- If the bank is the applicant, such a bank must belong to a nation whose central bank must be a member of the Bank for International Settlements.

Check Your Progress 4

- 1) Examine the numerous advantages of Foreign Portfolio Investment (FPI).

.....

.....

.....

.....

- 2) Examine the various constraints and drawbacks associated with Foreign Portfolio Investment (FPI).

.....

.....

.....

.....

- 3) Describe the prerequisites for qualifying as a Foreign Portfolio Investor (FPI).

.....

.....

.....

.....

23.13 LET US SUM UP

This unit was concerned with foreign investment. Two types of foreign investment are considered: foreign direct investment and foreign portfolio investment. The unit discussed the concept of each and also brought out the differences between the two.

We saw that FDI is an important source of external private finance. The unit explained that FDI is a way of integrating international economics as it creates and promotes stable and long-lasting links between economies. It offers the following major benefits to economies: a) promotes international trade through access to foreign markets, b) facilitates the transfer of technology between countries, c) leads to economic development, and d) creates job opportunities. The unit carefully explained the concept of FDI, You were familiarised with the methods of FDI like Greenfield investment, mergers and acquisitions joint ventures and subsidiary arrangements. The unit further described the types of FDI and spelt out the advantages and disadvantages of FDI.

The unit then went on to discuss FPI. We saw that Foreign Portfolio Investment (FPI) is very important in modern economies that have sophisticated financial systems. FPI allows foreign parties to invest in financial assets and instruments in the host country. This has a positive bearing on the price of domestic assets. The unit explained the types of FPI and the difference between FDI and FPI. Following this the unit described the categories of FPI. The advantages and disadvantages of FPI were discussed. Finally, the unit discussed the eligibility criteria for FPI in India.

23.14 KEY WORDS

Automatic Route	:	It refers to the entry route through which investment by a person resident outside India does not require the prior approval of the Reserve Bank of India or the Central Government.
FPI	:	Foreign Portfolio Investments (FPI) refer to investments that are made in securities and other financial assets issued in another country.
Foreign Portfolio Investor	:	Foreign Portfolio Investor' (FPI) means a person who is registered by the provisions of the Securities and Exchange Board of India (Foreign Portfolio Investors) Regulations, 2019, as amended from time to time.
Government Route	:	Government Route refers to the entry route through which investment by a person resident outside India requires prior Government approval and foreign investment received under this route shall be by the conditions stipulated by the Government in its approval.

NSDL	:	National Securities Depository Limited. It facilitates monitoring of overall foreign investment limits as well as sub-limits that apply to NRI and FPI.
RBI	:	Reserve Bank of India. It plays an important role in regulating FDI in India. It provides for certain rules and regulations for FDI.
SEBI	:	Securities Exchange Board of India. SEBI provides guidelines for investments made in securities and other financial assets issued in another country.
NSDL	:	National Securities Depository Limited.
Foreign Portfolio Investor	:	Foreign Portfolio Investor' (FPI) means a person registered by the provisions of the Securities and Exchange Board of India (Foreign Portfolio Investors) Regulations, 2019, as amended from time to time.
Government Route	:	'Government Route' means the entry route through which investment by a person resident outside India requires prior Government approval and foreign investment received under this route shall be by the conditions stipulated by the Government in its approval.

23.15 SOME USEFUL BOOKS

- 1) Taxmann's FEMA & FDI Ready Reckoner with Guide to Overseas Investment, 18th Edition 2023
- 2) Foreign Direct Investment Statistics: How Countries Measure FDI 2001. (2003). United States: International Monetary Fund.

23.16 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Foreign direct investment (FDI) can be defined as an investment made outside the home country where an investor resident in one country invests in another country in a way that establishes a long interest in and a high degree of influence over an enterprise.
- 2) Methods of Foreign Direct Investment (FDI): Greenfield Investment, Mergers and Acquisitions (M&A), Joint Ventures, Subsidiary Companies.
- 3) Various types of FDI are: Horizontal FDI, Vertical FDI, Conglomerate FDI, Platform FDI.

Check Your Progress 2

- 1) Routes of FDI: Automatic Route, Government Route
- 2) Advantages of FDI: FDI contributes significantly to the economic growth of the host country, FDI creates new jobs in the host country as foreign companies and foreign investors bring advanced technology, expertise, and management practices, It leads to an increase in exports from the host country as foreign companies, to attract foreign investments in the country etc.
- 3) Disadvantages of FDI: FDI may result in a loss of control over their domestic industries, a country may become vulnerable to economic fluctuations leading to economic instability, Foreign investors often take profits back to their home countries, FDI even results in environmental degradation as foreign investors etc.

Check Your Progress 3

- 1) FPI: FPI means any investment made by a person resident outside India through equity instruments where such investment is less than 10% of the post-issue paid-up share capital on a fully diluted basis of a listed Indian company or less than 10% of the paid-up value of each series of an equity instrument of a listed Indian company.
- 2) Differentiate between Foreign Direct Investments and Foreign Portfolio Investments: In FDI a direct business interest is established overseas whereas FPI is not a direct investment, FPI involves passive investors, and FDI is all about active investors etc.
- 3) Various categories of Foreign Portfolio Investment: There are three categories: Category I: This category is regarded as a low-risk category, Category II: This category involves a medium degree of risk, and Category III: This category is the riskiest among all.

Check Your Progress 4

- 1) Advantages of Foreign Portfolio Investment: It stimulates demand for companies' stocks, offers greater diversification to investors, benefits from fluctuations in exchange rates etc.
- 2) The various constraints and drawbacks associated with Foreign Portfolio Investment (FPI) are that it may destabilize financial markets and exchange rates, may lead to speculative behaviour in financial markets, may expose a country's currency to exchange rate fluctuations etc.
- 3) Prerequisites for qualifying as a Foreign Portfolio Investor (FPI): must be a non-resident Indian, must not be a citizen of a country that has been categorised under the public statement of FATF, if the bank is the applicant, such a bank must belong to a nation whose central bank must be a member of the Bank for International Settlements etc.